Sedenion extension loops and frames of hypercomplex 2^n-ons

by

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DEDICATION

I would like to dedicate this thesis to my wife Jennifer and to my children Kevin and Edna who sacrificed their time to be separated from me as I undertook this work. I would also like to thank my parents for their great encouragement over the years in my quest for knowledge. Thanks to my friends and family for their loving guidance and prayers during the writing of this work.

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ABSTRACT

This research work is mainly on hypercomplex numbers with major emphasis on the 16 dimensional sedenions. In Chapter 1, we give the literature review on current and past work on the area of hypercomplex numbers. Basic definitions are given and the sedenions are introduced. The advantages and disadvantages of the different doubling formulas are discussed. In Chapter 2, we introduce sedenion extensions and give equivalent conditions for the extensions to be groups. Consequences arising from these conditions are also given. Chapter 3 gives a powerful result, the multiplication of the frames of the 2^n-ons fits in the projective geometry $PG(n-1,2)$. A formula for counting the number of multiplicative subloops of order 2^k inside the frames is given. In Chapter 4, Hadamard matrices are discussed. It is shown that the sign matrix of the frame multiplication in the 2^n-ons under the Smith-Conway or Cayley-Dickson process is a skew Hadamard matrix. These matrices are shown to be equivalent to Kronecker products when $n \leq 3$. Chapter 5 gives some open problems for future research.

CHAPTER 1. LITERATURE REVIEW AND BASIC DEFINITIONS

In this Chapter we give some review on "hypercomplex" numbers. We introduce the sedenions and the different doubling formulas for obtaining sedenions from the octonions and general 2^n-ons from 2^{n-1}-ons. We also discuss some advantages and disadvantages of these doubling formulas.

1.1 Introduction

In 1831, Gauss was convinced that there were no "hypercomplex" number system outside the complex numbers in which the basic properties of the complex numbers persisted. In the 1880's, a friendly dispute arose between Weierstrass and Dedekind about the proper interpretation of Gauss' words. The controversy revolved around the question of characterizing all finite dimensional commutative and associative real algebras with a unit element, zero divisors allowed.

In 1843, Sir William Hamilton discovered the quaternions and shortly afterwards Graves and Cayley constructed the octonions. A more detailed analysis of the quaternions can be found [9, Chap. 7]. In 1877, Frobenius showed that any non-zero associative, quadratic real algebra without zero divisors is isomorphic to the real, complex or the quaternions. The Gelfand-Mazur Theorem establishes that every commutative real division algebra $\mathcal{A} \neq 0$ is isomorphic to $\mathbb{R}$ or $\mathbb{C}$ [9, cf p. 229-243].

It was not the discovery of the quaternions which was the great achievement, but rather the recognition which came as a result of the discovery; the great freedom to construct hypercomplex systems. The non-zero octonions form a loop and can be looked at from the point of view of loop theory.

The area of loop theory has attracted great interest in recent years and many mathematicians have paid great attention to it in the 21st century.

Definition 1.1.1 A groupoid $\langle G, \cdot \rangle$ is a non-empty set which is closed under the binary operation, i.e, for all $a, b \in G$ there exists a unique $c \in G$ such that $a \cdot b = c$. Semigroups are groupoids satisfying the associative property. A *quasigroup* is a groupoid such that in the equation $x \cdot y = z$, knowledge of any two of the elements defines the third one uniquely.

We can define left and right multiplications in a quasigroup Q by:

$$\text{Left multiplication:} \quad L(x) : Q \to Q, \; q \to xq \tag{1.1}$$

$$\text{Right multiplication:} \quad R(x) : Q \to Q, \; q \to qx \tag{1.2}$$

where the multiplication is a bijection for all $x \in Q$. A quasigroup can also be defined as a set Q with binary operation $\cdot$, left division $\backslash$ and right division $/$ such that the identities

$$\begin{aligned} x\backslash(xy) &= y \\ x \cdot (x\backslash y) &= y \end{aligned} \tag{1.3}$$

and

$$\begin{aligned} (y \cdot x)/x &= y \\ (y/x) \cdot x &= y \end{aligned} \tag{1.4}$$

are satisfied. In this aspect a *left quasigroup* is a set Q satisfying (1.3) while a *right quasigroup* satisfies (1.4). A *loop* L is a quasigroup with a two-sided identity e satisfying

$$e \cdot x = x = x \cdot e, \quad \forall\, x \in Q \tag{1.5}$$

while a *left\right loop* has a left\ resp. right identity satisfying

$$x \backslash x = x \backslash (x \cdot e) = e \quad \forall\, x \in Q \tag{1.6}$$

respectively

$$x/x = (e \cdot x)/x = e \quad \forall\, x \in Q \tag{1.7}$$

A *subloop* of a loop L is a nonempty subset B of L with $B \neq L$ which is itself a loop. In the language of loop theory, groups are simply associative loops. A loop L has *left\right inverse property* if (1.8) (resp. (1.9)) is satisfied.

$$\forall\, x \in L,\ \exists\, x^l \in L \quad \text{such that} \quad x^l(x \cdot y) = y,\ \forall\, y \in L \tag{1.8}$$

$$\forall\, x \in L,\ \exists\, x^r \in L \quad \text{such that} \quad (y \cdot x)x^r = y,\ \forall\, y \in L \tag{1.9}$$

It has the *inverse property* if it has both left and right inverse properties and

$$x \cdot x^r = x^l \cdot x = e,\ \forall\, x \in L.$$

It is said to be *power associative* if the subloops generated by single elements are groups. It is *diassociative* if the subloops generated by any pairs of elements are groups. Power associativity implies two-sided identities under the involution map

$$x \to x^{-1} \quad \text{with} \quad x \cdot x^{-1} = e = x^{-1} \cdot x$$

while diassociativity implies both the right and the left inverse properties since

$$y = y(xx^{-1}) = yx \cdot x^{-1} \quad \text{and} \quad x^{-1}x \cdot y = x^{-1} \cdot xy = y.$$

In order to study a wide variety of loops, one studies loops satisfying some weak form of the associativity law. Three elements $x,\ y,\ z$ in a loop L *associate* if $xy \cdot z = x \cdot yz$. A loop L is called a *Moufang loop* if the Moufang identities

$$xy \cdot zx = (x \cdot yz)x \tag{1.10}$$

$$x(y \cdot xz) = (xy \cdot x)z \tag{1.11}$$

$$x(y \cdot zy) = (x \cdot yz)y \tag{1.12}$$

are satisfied for every $x,\ y,\ z \in L$. It is a *proper* Moufang loop if it is Moufang but not associative. The non-zero octonions form a proper Moufang loop. Any one of the three identities implies the other two (cf. [2, p.78]). In a Moufang loop, if three elements associate, they associate in any order as guaranteed by the the crucial *Moufang Theorem* first proved by R. Moufang [13].

Theorem 1.1.2 (Moufang Theorem) *Let $x,\ y$ and z be elements in a Moufang loop L. Then the subloop generated by $x,\ y$ and z is associative if and only if $x,\ y$ and z associate.*

Moufang loops are power associative and diassociative by the Moufang Theorem. Moreover, every element has a unique two-sided inverse. A *Bol loop* is a loop satisfying the left or right "Bol laws"

$$\text{Left Bol law:} \quad (x \cdot yx)z = x(y \cdot xz) \tag{1.13}$$

$$\text{Right Bol law:} \quad z(xy \cdot x) = (zx \cdot y)x \tag{1.14}$$

We have a left or right Bol if the corresponding laws are satisfied. If both are satisfied, the loop is Moufang. The following table gives some commonly used definitions characterizing properties in loops. We consider a loop L with identity $e,\ x,\ y \in L$, and scalars $\alpha,\ \beta$.

Table 1.1 Characterization of loop properties

Property	**Characterization**
Antiautomorphism	$\overline{xy} = \overline{y}\,\overline{x}$
Power distributive	$(\alpha x^k + \beta x^m)y = \alpha x^k y + \beta x^m y$
Right alternative	$x \cdot y^2 = xy \cdot y$
Left alternative	$x \cdot xy = x^2 \cdot y$
Weak linearity	$(x + \alpha e)y = xy + \alpha y,\ y(x + \alpha e) = yx + \alpha y$
Right linear	$x(y + z) = xy + xz$
Left linear	$(x + y)z = xz + yz$
Flexible	$xy \cdot x = x \cdot yx$
Scaling	$\alpha xy = \alpha x \cdot y = x \cdot \alpha y$
Jordan	$x^2 y \cdot x = x^2 \cdot yx,\ xy \cdot x^2 = x(y \cdot x^2)$

It is worth noting that if the multiplication in a loop is flexible, the distinction between Bol and Moufang identities vanish. The loop is *abelian* if it is both commutative and associative.

1.2 The Sedenions

The complex numbers $\mathbb{C}$, quaternions $\mathbb{H}$, and the Cayley numbers $\mathbb{K}$ are real division algebras obtained from the real numbers through successive doubling. The "Cayley-Dickson process" has been in use for a long time. On doubling the Cayley numbers, we obtain the *Cayley-Dickson sedenions* denoted by $\mathbb{S}$. The word sedenion comes from the Latin word *sedecim* meaning 16. In each step of the doubling process, there is a continuous degradation of the algebraic properties. The complex numbers are no longer ordered and the self conjugate property is lost. The quaternions are not commutative while the associative property is lost in the Cayley numbers.

The associative algebras may be described easily as algebras of real matrices, the complex numbers can be viewed as matrices of the form;

$$x + iy = \begin{bmatrix} x & -y \\ y & x \end{bmatrix} \quad \text{where } x, y \in \mathbb{R}$$

and the quaternions viewed as matrices of the form;

$$u + vj = \begin{bmatrix} u & -v \\ \overline{v} & \overline{u} \end{bmatrix} \quad \text{where } u, v \in \mathbb{C}.$$

By writing $u = t + xi,\ v = y + zi$ and using the complex matrix representation, we get;

$$t + xi + yj + zk = \begin{bmatrix} t & -x & -y & z \\ x & t & -z & -y \\ y & z & t & x \\ -z & y & -x & t \end{bmatrix} \quad \text{where } t, x, y, z \in \mathbb{R}$$

In [23], the sedenions are represented by $\mathbb{S} = \mathbb{K} \oplus \mathbb{K}f$ and the Cayley numbers appear as a subalgebra of $\mathbb{S}$. The right distributive law is lost although the left distributive law still holds.

R. Cawagas [4] has been studying the sedenions arising from the Cayley-Dickson doubling process. The name "Cayley-Dickson sedenions" is used to avoid ambiguity with those arising from other doubling formulations. The sedenion frame under the Cayley-Dickson process has 67 subloops, 66 nontrivial, 15 of order 16 where 8 are octonion loops and 7 are quasi-octonion loops. There are 35 quaternion groups, 15 cyclic groups of order 4 (isomorphic to complex numbers), one of order 2 and the trivial group. The subloops are fully characterized in [4]. J.D Phillips and Petr [20] have introduced varieties of loops of Bol-Moufang which satisfy the identity $(xy \cdot x)z = x(y \cdot xz)$ and showed that there are exactly 14 such varieties.

Several scholars have come up with different doubling formulas with an interest of getting nicer properties in the sedenions. We here give some of the different doubling formulas, together with their advantages and disadvantages.

1.3 Comparison of doubling formulas

The earliest known and most natural method is the Cayley-Dickson doubling process given by:

$$(a, b)(c, d) = (ac - \overline{d}b, da + b\overline{c}) \tag{1.15}$$

The formula applied to the Cayley numbers gives the Cayley-Dickson sedenions. The multiplication is power associative and satisfies the quadratic, flexible, antiautomorphism and Jordan identities. However, the sedenions obtained by this process include zero-divisors, the norm is not multiplicative and the multiplication is not alternative. S^{15}, "sphere" of unit length vectors in $\mathbb{S}$, is not closed under multiplication while the identity element is not two sided. All Cayley-Dickson algebras are strictly power associative since all commutative Jordan algebras are, Bremner and Hentzel [3].

Zassenhaus and Eichhorn found a rational 16-square identity based on defining the determinant of a 2×2 matrix whose entries are octonions. The multiplication formula is:

$$(a, b)(c, d) = (a\overline{c} - b\overline{d},\ ad + ac \cdot a^{-1}b) \tag{1.16}$$

This multiplication is not power associative and has no left identity. It is only left linear, for the form $(a, 0)$ and does not obey weak-linearity and quadratic identity. However, it is right linear, obeys real-part antiautomorphism $re(\overline{ab}) = re(\overline{b}\ \overline{a})$ and the Jordan identity holds in coordinates 1-15.

Conway and Derek Smith [6, p.79], gave a modification of the Cayley-Dickson process defined by:

$$(a, b)(c, d) = \begin{cases} (ac,\ \overline{a}d) & \text{if } \ b = 0 \\ (ac - \overline{\overline{b}d},\ b\overline{c} + b(\overline{\overline{a}.\overline{b^{-1}d}})) & \text{if } \ b \neq 0. \end{cases} \tag{1.17}$$

Warren D. Smith and Conway (private communication) have come up with a new and more recent approach based on matrix multiplications. The multiplication is given by

$$(a,b)(c,d) = \begin{cases} (ac,\ \overline{a}d) & \text{if} \quad b = 0 \\ (ac - \overline{b\overline{d}},\ \overline{\overline{b}\,\overline{c}} + \overline{\overline{b}\,\overline{a}\ \overline{\overline{\overline{b^{-1}}}\ \overline{d}}}) & \text{if} \quad b \neq 0. \end{cases} \tag{1.18}$$

The multiplication is power associative, right linear, power distributive and has antiautomorphism property. It is left linear only for the first nine coordinates and does not obey any Bol identities. In fact, it obeys the left Bol identity in real coordinates only, $re([x.yx])z = re(x[y.xz])$. There are no two-sided loops since the *niners*, sedenions of the form (a,b) where $b \in \mathbb{R}$ and $a \in \mathbb{K}$, are not closed under multiplication.

Jonathan Smith [23, p.132], introduced the formula:

$$(a,b)(c,d) = \begin{cases} (ac,\ da) & \text{if} \quad b = 0 \\ (ab \cdot cb^{-1} - b\overline{d},\ b\overline{c} + db^{-1} \cdot ab) & \text{if} \quad b \neq 0 \end{cases} \tag{1.19}$$

This multiplication is not power associative, not left or right alternative, and neither weak nor left linear. The quadratic and Jordan identities are not satisfied. It is right linear and $\mathbb{K}$ forms a subalgebra of $\mathbb{S}$.

This multiplication gives rise to the two-sided loops namely *Sedenion Loop Extensions*, denoted by $L \rtimes S^0$ which are niners (a,b) with $b \in S^0 = \{1,-1\}$ and L a subloop of $\mathbb{K}^*$. In Chapter 2, we investigate the properties of these extensions. Here the extensions are constructed from explicit subloops L of $\mathbb{K}^*$ or just S^7.

CHAPTER 2. SEDENION EXTENSION LOOPS

This chapter investigate sedenion multiplication from the standpoint of loop theory. Sedenion extensions, which are two-sided loops, are obtained within the version of the sedenions introduced by Jonathan Smith [23]. Conditions are given for the satisfaction of standard loop-theoretical identities within these loops. The equivalent relations for the sedenion extension to be a group are given with their consequences.

2.1 Introduction

One of the most mature parts of loop theory is the theory of Moufang loops. Moufang loops trace their origins back to the primal examples given by various loops of non-zero or unit-norm octonions. The octonions furnish algebra structure on 8-dimensional Euclidean space, the Euclidean norm $|x|$ being *multiplicative* in the sense that:

$$|xy| = |x| \cdot |y| \tag{2.1}$$

under octonion multiplication. For n-dimensional Euclidean space, multiplicativity of the norm reduces to a question that has long been of interest to many mathematicians:

Can the product of two sums of n squares be expressed as a sum of n squares?

In other words:

$$(a_1^2 + a_2^2 + \cdots + a_n^2)(b_1^2 + b_2^2 + \cdots + b_n^2) = A_1^2 + A_2^2 + \cdots + A_n^2. \tag{2.2}$$

A famous 1898 theorem of Hurwitz [6] [9] shows that an algebra (i.e, with both left and right distributivity) has a multiplicative Euclidean norm only for $n = 1, 2, 4$ and 8. In particular, the usual Cayley-Dickson process for starting from the reals $\mathbb{R}$ and successively generating the complex numbers $\mathbb{C}$, Hamilton's quaternions $\mathbb{H}$, and the octonions $\mathbb{K}$, each with multiplicative Euclidean norm, does not make the Euclidean norm on 16-dimensional space multiplicative. In 1967, Pfister [21] proved that for $n = 2^k$, a product may be defined on Euclidean n-space such that (2.2) holds. However, Pfister was not concerned with algebraic properties of the product. Subsequently, there have been various multiplications defined on the 16-dimensional real space with the intention of rendering the Euclidean norm multiplicative (cf. [6§6.11], [27]).

In [23, p.132], the formula in Equation (1.19) for multiplication on 16-dimensional Euclidean space $\mathbb{K} \oplus \mathbb{K}f$ was introduced. This multiplication does make the Euclidean norm multiplicative and the octonions $\mathbb{K}$ form a subalgebra of $\mathbb{S}$. The non-zero sedenions form a left loop under multiplication (in the strong sense [24, Ch. I, §4.3] that includes a two-sided identity).

The intention of the present chapter is to consider the sedenions from the point of view of loop theory. Although the sedenions do not directly yield loops in the way that Moufang loops are obtained from octonions, the left loop of non-zero sedenions does contain new two-sided subloops defined in Section 2.2 below as sedenion extensions. These loops are constructed abstractly as extensions of subloops L of the octonions. The chapter examines the satisfaction by these extensions of various standard loop-theoretical identities. For example, it turns out that they are all power-associative (Corollary 2.4.2), even though the full left loop of all non-zero sedenions is not itself power-associative. In this Chapter, the sedenions take the form $\mathbb{S} = \mathbb{K} + \mathbb{K}f$ as in [23].

2.2 Sedenion extensions

Let L be a multiplicative subloop of the non-zero octonions. Then its *sedenion extension* $L \rtimes S^0$ is the disjoint union $L \cup Lf$ within the sedenions. Elements of this union are encoded as pairs (a, ε), with $\varepsilon \in S^0 = \{1, -1\}$, by:

$$a \mapsto (a, 1), \quad af \mapsto (a, -1)$$

[23, (5.4)]. Specializing (1.19), in the full sedenion extension $\mathbb{K}^* \rtimes S^0$ of the loop $\mathbb{K}^*$ of all non-zero octonions, the multiplication is given by:

1. $(x, 1)(y, 1) = (xy, 1)$
2. $(x, 1)(y, -1) = (yx, -1)$
3. $(x, -1)(y, 1) = (x\overline{y}, -1)$
4. $(x, -1)(y, -1) = (-x\overline{y}, 1)$

[23, Prop. 5.2]. The sedenion extensions are certainly two-sided loops, with right divisions given by:

1. $(x, 1)/(y, 1) = (x/y, 1)$
2. $(x, 1)/(y, -1) = (-x/\overline{y}, -1)$
3. $(x, -1)/(y, 1) = (x/\overline{y}, -1)$
4. $(x, -1)/(y, -1) = (y\backslash x, 1)$.

Consider a sequence (finite or infinite) of groups and homomorphisms

$$\cdots \longrightarrow A_1 \xrightarrow{\phi_1} A_2 \xrightarrow{\phi_2} A_3 \longrightarrow \cdots$$

the sequence is said to be *exact at* A_2 if image $\phi_1 =$ kerl ϕ_2. If it is everywhere exact, it is said to be an *exact sequence.* If the sequence

$$1 \longrightarrow A_1 \xrightarrow{\phi} A_2 \xrightarrow{\psi} A_3 \longrightarrow 1$$

is exact, such a sequence is called a *short exact sequence.* Then $A_2/\phi(A_1)$ is isomorphic to A_3; the isomorphism is induced by ψ. The sequence is said to *split* if $\phi(A_1)$ is a direct summand in A_2. This means that A_2 is the internal direct sum of $\phi(A_1)$ and some other subgroup B; the group B is not uniquely determined. The sequence becomes

$$1 \longrightarrow A_1 \xrightarrow{\phi} \phi(A_1) \oplus B \xrightarrow{\psi} A_3 \longrightarrow 1$$

where ϕ defines an isomorphism of A_1 with $\phi(A_1)$, and ψ defines an isomorphism of B with A_3. These definitions of exactness are extended to loops.

Each sedenion extension $L \rtimes S^0$ appears in an exact sequence:

$$1 \longrightarrow L \xrightarrow{j} L \rtimes S^0 \xrightarrow{p} S^0 \longrightarrow 1 \tag{2.3}$$

of loops with $j : x \mapsto (x, 1)$ and $p : (x, \varepsilon) \mapsto \varepsilon$.

2.3 The main theorem

We begin with a preliminary observation required for the proof of the main theorem. Recall that a loop is said to be *abelian* if it is both commutative and associative, i.e, an abelian group.

Lemma 2.3.1 *Each commutative multiplicative subloop of the octonions is associative.*

Proof.

A commutative multiplicative subloop L of the octonions spans a commutative subalgebra R of the octonions. This subalgebra R is a commutative, alternative division ring.

As such, it is associative [2, Lemma 6, p. 133][1]. Thus, its subreduct L is also associative.

□

Theorem 2.3.2 *Let L be a multiplicative subloop of the octonions. Then the following statements are equivalent:*

1. *$L \rtimes S^0$ is a group;*
2. *$L \rtimes S^0$ has the left inverse property;*
3. *$L \rtimes S^0$ has the right inverse property;*
4. *L is commutative;*
5. *L is abelian.*

Proof.

- The implications (1) $\Rightarrow$ (2) and (1) $\Rightarrow$ (3) are immediate.
- (2) $\Rightarrow$ (4): Consider an element x of L. Note that

$$(-x/|x|^2, -1)(x, -1) = (x\overline{x}/|x|^2, 1) = (1, 1). \tag{2.4}$$

Then, for a further element y of L,

$$\begin{aligned}(-x/|x|^2, -1)\left[(x, -1)(y, 1)\right] &= (-x/|x|^2, -1)(x\overline{y}, -1) \\ &= (xy\overline{x}/|x|^2, 1) = (xyx^{-1}, 1).\end{aligned} \tag{2.5}$$

If the left inverse property holds in $L \rtimes S^0$, the L-component of the final term in (2.5) has to be y. Thus, L is commutative.

- (3) $\Rightarrow$ (4) is proved in dual fashion to the previous implication. First, note directly that

$$(x, -1)(-x/|x|^2, -1) = (x\overline{x}/|x|^2, 1) = (1, 1) \tag{2.6}$$

[1]If R is a commutative subloop of the octonions, then it is associative and thus a field.

for an element x of L. Once Corollary 2.4.2 is established, equation (2.6) follows immediately from (2.4). Then, for a further element y of L,

$$\begin{aligned}[(y,1)(x,-1)]\,(-x/|x|^2,-1) &= (xy,-1)(-x/|x|^2,-1) \\ &= (xy\overline{x}/|x|^2,1) = (xyx^{-1},1).\end{aligned} \tag{2.7}$$

If the right inverse property holds in $L \rtimes S^0$, the L-component of the final term in (2.7) has to be y. Thus, again L is commutative.

- (4) $\Rightarrow$ (5): We apply Lemma 2.3.1.

- (5) $\Rightarrow$ (1): Suppose that L is abelian. There are seven non-trivial cases of associativity to consider in $L \rtimes S^0$, arising whenever an S^0-component ε, ζ, η in the following equation is negative:

$$(x,\varepsilon)\,[(y,\zeta)(z,\eta)] = [(x,\varepsilon)(y,\zeta)](z,\eta).$$

The computations for these cases are as follows:

$$\begin{aligned}[(x,1)(y,1)]\,(z,-1) &= (xy,1)(z,-1) = (zxy,-1) \\ &= (zyx,-1) = (x,1)(zy,-1) = (x,1)[(y,1)(z,-1)]\end{aligned} \tag{2.8}$$

$$\begin{aligned}[(x,1)(y,-1)](z,1) &= (yx,-1)(z,1) = (yx\overline{z},-1) \\ &= (y\overline{z}x,-1) = (x,1)(y\overline{z},-1) = (x,1)[(y,-1)(z,1)]\end{aligned} \tag{2.9}$$

$$\begin{aligned}[(x,1)(y,-1)](z,-1) &= (yx,-1)(z,-1) = (-yx\overline{z},1) \\ &= (-xy\overline{z},1) = (x,1)(-y\overline{z},1) = (x,1)[(y,-1)(z,-1)]\end{aligned} \tag{2.10}$$

$$\begin{aligned}[(x,-1)(y,1)](z,1) &= (x\overline{y},-1)(z,1) = (x\overline{y}\;\overline{z},-1) \\ &= (x(\overline{yz}),-1) = (x,-1)(yz,1) = (x,-1)[(y,1)(z,1)]\end{aligned} \tag{2.11}$$

$$[(x,-1)(y,1)](z,-1) = (x\overline{y},-1)(z,-1) = (-x\overline{y}\ \overline{z},1) \tag{2.12}$$
$$= (-x(\overline{z}\overline{y}),1) = (x,-1)(zy,-1) = (x,-1)[(y,1)(z,-1)]$$

$$[(x,-1)(y,-1)](z,1) = (-x\overline{y},1)(z,1) = (-x\overline{y}\ z,1) \tag{2.13}$$
$$= (-x(\overline{y\overline{z}}),1) = (x,-1)(y\overline{z},-1) = (x,-1)[(y,-1)(z,1)]$$

$$[(x,-1)(y,-1)](z,-1) = (-x\overline{y},1)(z,-1) = (-zx\overline{y},-1) \tag{2.14}$$
$$= (-x(\overline{y\overline{z}}),-1) = (x,-1)(-y\overline{z},1) = (x,-1)[(y,-1)(z,-1)].$$

□

2.4 Consequences of the theorem

The first consequence of Theoren 2.3.2 is negative.

Corollary 2.4.1 *No sedenion extension loop can be a proper Moufang or Bol loop.*

Proof.

Moufang loops satisfy both inverse properties, while left or right Bol loops satisfy the respective left or right inverse property [24, Ch.I, §§4.1–2]. Thus, a sedenion extension satisfying a Moufang or Bol law would be a group. □

On the positive side, the following result contrasts with the observation that the full left loop of non-zero sedenions under Equation (1.19) is not power-associative [23, Prop. 5.1(i)].

Corollary 2.4.2 *Each sedenion extension loop is power-associative.*

Proof.

Each element (x,ε) of a sedenion extension loop is an element of the sedenion extension of the abelian group $\langle x\rangle$. By Theorem 2.3.2, this extension is associative. □

Since each sedenion extension loop $L \rtimes S^0$ is power-associative, it possesses an *exponent* defined in the classical group-theoretical way as either 0 or else the smallest positive integer n for which $L \rtimes S^0$ satisfies the identity $x^n = 1$. (Recall that a related concept of exponent for a general quasigroup in a variety of quasigroups was defined in [22, Defn. 5.2].) The next result shows how the exponent of a multiplicative subloop of the octonions determines the exponent of its sedenion extension.

Proposition 2.4.3 *Let L be a multiplicative subloop of the octonions.*

1. *Suppose L has positive exponent n. Then the sedenion extension $L \rtimes S^0$ has a positive exponent m which is the least common multiple of* 4 *and n,* $\mathrm{LCM}\{4, n\}$.

2. *Suppose L has exponent* 0*. Then the sedenion extension $L \rtimes S^0$ has exponent* 0.

Proof.

1. Suppose $L \rtimes S^0$ satisfies $x^k = 1$. First note $(1, -1)^2 = (-1, 1) \neq (1, 1)$ and $(1, -1)^4 = (1, 1)$, so $4|k$. Also $n|k$, since L is a subloop of $L \rtimes S^0$. In other words, the exponent of $L \rtimes S^0$ is either zero or a multiple of m. Conversely, consider $a \in L$. Then $a^n = 1 \Rightarrow a^m = 1 \Rightarrow |a|^m = 1$. Now, $(a, 1)^m = (a^m, 1) = (1, 1)$. Also, $(a, -1)^m = (|a|^m, 1)$ since $4|m$, so $(a, -1)^m = (1, 1)$. Thus, $L \rtimes S^0$ does satisfy $x^m = 1$.

2. If L has elements of infinite order, then so does $L \rtimes S^0$ since it contains L as a subloop.

$\square$

Remark 2.4.4 Note that since a sedenion extension $L \rtimes S^0$ always contains an element $(1, -1)$ of exponent 4, it is never torsion-free. This contrasts with the conjecture of [6,

§7.4] that "the loop generated by n generic real octonions is in fact the free Moufang loop on n generators."

The final considerations concern commutativity of sedenion extensions.

Proposition 2.4.5 *Let L be a multiplicative subloop of the octonions. Then the sedenion extension $L \rtimes S^0$ is commutative if and only if L is a multiplicative subgroup of the reals.*

Proof.

If x and y are elements of a multiplicative subgroup L of the reals, then

$$(x,1)(y,-1) = (yx,-1) = (y\overline{x},-1) = (y,-1)(x,1)$$

and

$$(x,-1)(y,-1) = (-x\overline{y},1) = (-xy,1) = (-y\overline{x},1) = (y,-1)(x,-1);$$

so the sedenion extension is commutative. Conversely, if x is an element of a commutative sedenion extension $L \rtimes S^0$, then

$$(-\overline{x},1) = (1,-1)(x,-1) = (x,-1)(1,-1) = (-x,-1);$$

so x is real. $\square$

To conclude, we return to the starting point and obtain an analogue of Lemma 2.3.1.

Corollary 2.4.6 *A commutative sedenion extension loop is associative.*

Proof.

Suppose that a sedenion extension loop $L \rtimes S^0$ is commutative. By Proposition 2.4.5, L is a subgroup of the reals, and so abelian. The extension $L \rtimes S^0$ then satisfies the equivalent conditions (1)–(3) of Theorem 2.3.2. In particular, it is associative. $\square$

Remark 2.4.7 Corollary 2.4.6 would be a more complete analogue of Lemma 2.3.1 if it were known that the only two-sided subloops of the multiplicative left loop of non-zero sedenions appear either as subloops of the octonions, or as sedenion extensions in the sense of this chapter. At present, however, this problem is open.

CHAPTER 3. PROJECTIVE GEOMETRY AND FRAME MULTIPLICATION IN 2^n-ONS

In this Chapter, we show that the frame multiplication of the 2^n-ons can be fit into the projective geometry $PG(n-1, 2)$. We show that the multiplication can be viewed in terms of designs, Nim-addition and projective geometry. A counting procedure for the number of subloops inside the frame multiplication is given.

3.1 Introduction

In 1892, Fano described an n-dimensional geometry in which the number of points on each line is $p+1$, p prime. In 1906, O. Veblen and W.H. Bussey gave this finite projective geometry the name $PG(n,p)$ and extended to $PG(n,q)$, $q = p^k$ [7, pg. 91]. Smith, Conway [6] and many others have characterized the multiplication of the basic elements of the octonions (Cayley numbers). They have shown that the multiplication can be fit in a Fano plane, the projective space PG(2,2). The basic elements corresponding to each line, together with 1, linearly generate a subalgebra isomorphic to the quaternions. In this chapter, we show that the multiplication of the basic elements of the sedenions under the Smith-Conway doubling processes Equations (1.17) and (1.18) can be fit into the three-dimensional projective space PG(3,2). Each of the 35 lines in the projective space corresponds to a triple of basic elements which, together with 1, linearly generate a subalgebra isomorphic to the quaternions, while each of the 15 Fano planes corresponds to a septuple of basic elements which, together with 1, linearly generates a subalgebra isomorphic to the Cayley numbers. More generally, we show that the multiplication of the basic elements of the 2^n-ons fits in to the $PG(n-1,2)$ projective geometry. In doing so, it is convenient to consider this projective geometry from several different viewpoints: as a design (Section 3.2), in terms of Nim addition (Section 3.5), and as the geometry of linear subspaces of a vector space (Section 3.7). It is then shown that the number of nontrivial subloops of order 2^k inside the loop formed by the *frame* of the 2^n-ons, the set of basic elements together with their negatives, is given by

$$\begin{bmatrix} n \\ k \end{bmatrix}_2 = \frac{(2^n-1)(2^{n-1}-1)(2^{n-2}-1)\cdots(2^{n-k+1}-1)}{(2^1-1)(2^2-1)(2^3-1)\cdots(2^k-1)}.$$

This extends results of Cawagas [4] on the sedenions to the general 2^n-ons.

3.2 Designs

Definition 3.2.1 Let $v,\kappa,\tau,\lambda \in \mathbb{Z}$ with $v \geq \kappa \geq \tau \geq 0$, $\lambda \geq 1$. A τ *-design* on v points with block size κ and index λ is an incidence structure $\mathcal{D} = (\mathcal{P},\mathcal{B},\mathcal{I})$ with

1. $|\mathcal{P}| = v$
2. $|B| = \kappa$ for every $B \in \mathcal{B}$
3. For each set T of τ points, there are exactly λ blocks incident with all such points in T.

Such a design is denoted by $\tau-(v,\kappa,\lambda)$ or $S_\lambda(\tau,v,\kappa)$. A *Steiner system* $S(\tau,\kappa,v)$ is a τ-design with $\lambda=1$. The special case $\tau=2,\ \kappa=3$ is called a *Steiner triple system.*

A Steiner triple system $S(2,3,v)$ of order v exists if and only if $v \equiv 1,3 \pmod 6$, Kirkman [17]. Against this background, the Fano plane arises as follows.

Definition 3.2.2 Take as points the elements of $\mathbb{Z}_7$ and as blocks all triples $B_x = \{x,\ x+1,\ x+3\}$ with $x \in \mathbb{Z}_7$. We obtain an $S(2,3,7)$, this design is the *Fano plane* as represented by Figure 3.1

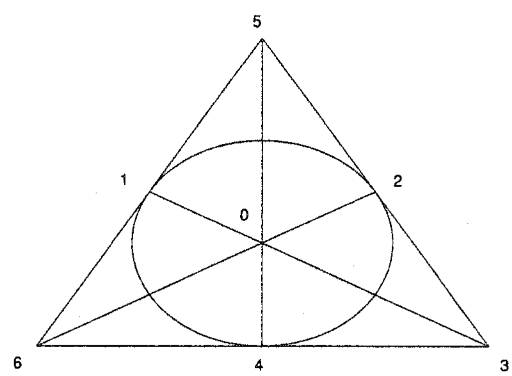

Figure 3.1 The Steiner triple system $S(2,3,7)$, the Fano plane

3.3 Multiplication of Octonion Frames

The Smith-Conway multiplications are given by Equations (1.17) and (1.18) and the two multiplications agree on the signed basic elements of the 2^n-ons, where they reduce to the form:

$$\begin{aligned}(a,0)(c,0) &= (ac,0)\\ (a,0)(0,d) &= (0,\bar{a}d)\\ (0,b)(c,0) &= (0,\bar{\bar{b}\bar{c}}) = (0,cb)\\ (0,b)(0,d) &= (-\bar{\bar{b}d},0) = -(d\bar{b},0).\end{aligned} \tag{3.1}$$

The basic elements of the octonions are $1, i_0, i_1, i_2, i_3, i_4, i_5, i_6$, where $1 = (1,0,0,0,0,0,0,0)$ and $i_k, 0 \le k \le 6$ has a 1 in the $k+2$ position and zeros elsewhere. The multiplication of these basic elements is characterized by combining $i_m, i_{m+1}, i_{m+3} \pmod 7$ and taking

$i_m = i$, $i_{m+1} = j$ and $i_{m+3} = k$. We characterize the multiplication of the signed elements of the sedenions and 2^n-ons. Here we use the Smith-Conway multiplication Equation (3.1).

Table 3.1 Octonion frame triplets

i	i_0	i_1	i_2	i_3	i_4	i_5	i_6
j	i_1	i_2	i_3	i_4	i_5	i_6	i_0
k	i_3	i_4	i_5	i_6	i_0	i_1	i_2

Table 3.1 gives the (i, j, k)-combinations of the signed elements of the octonions as characterized by Smith and Conway [6].

Each of the basic elements and their conjugates are inverses. The eight basic elements together with their negatives form a loop under octonion multiplication. The multiplications can be set up into a Fano plane as in Figure 3.1 with the seven basic elements, excluding the identity. Each line of the Fano plane represents an (i, j, k)-combination, and the corresponding basic elements together with 1 span a subalgebra of the Cayley numbers isomorphic to the quaternions.

3.4 Fano planes in the sedenion loop

From now on we will use a new numbering for the signed elements. This makes the multiplications and characterizations much easier, as will be realized in the sections that follow.

We take $B_{\mathbb{C}} = \{e_0 = (1,0),\ e_1 = i = (0,1)\}$ to be the basic elements of the complex numbers and

$$B_{\mathbb{Q}} = \{e_{00} = (e_0, 0), e_{01} = (e_1, 0), e_{10} = (0, e_0), e_{11} = (0, e_1)\}$$

as the basic elements of the quaternions, with multiplication given by $e_{01} \cdot e_{11} = e_{10}$.

The basic elements of the octonions are given by

$$B_{\mathbb{K}} = \{e_{000} = (e_{00}, 0),\ e_{001} = (e_{01}, 0),\ e_{010} = (e_{10}, 0),\ e_{011} = (e_{11}, 0),$$
$$e_{100} = (0, e_{00}), e_{101} = (0, e_{01}),\ e_{110} = (0, e_{10}),\ e_{111} = (0, e_{11})\}.$$

The 'non-identity' basic element multiplications can be fit into a new Fano plane, Figure 3.2. Each line in the Fano plane corresponds to a subalgebra isomorphic to the

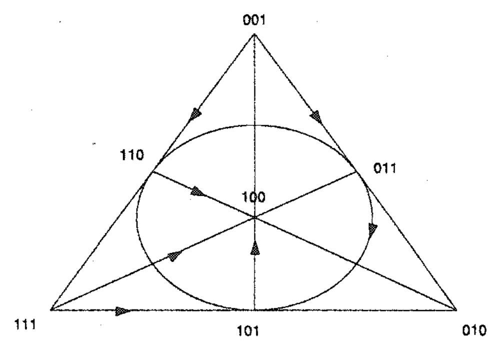

Figure 3.2 Fano plane for octonion frame multiplication

quaternions. There are a total of 7 lines, and thus 7 subalgebras isomorphic to the quaternions. For each triple $e_ie_j \in \{e_k,\ -e_k\}$, the arrows show the direction in which the multiplication is positive, while against the arrow gives a negative result.

The basic elements of the sedenions are given by:

$$B_{\mathbb{S}} = \{e_{0000}, e_{0001}, e_{0010}, \ldots, e_{1111}\}$$

where each element is of the form $(a, 0)$ or $(0, a)$ with $a \in B_{\mathbb{K}}$. Writing each basic element explicitly,

$$e_{0k} = (e_k, 0),\ e_{1k} = (0, e_k),\ e_k \in B_{\mathbb{K}} \text{ for } k \in \{000, 001, \ldots, 111\}.$$

The following properties will be used in making the multiplication table for the signed basic elements of the sedenions.

Properties 3.4.1

1. e_{0000} is the two-sided identity since for all $a \in \mathbb{K}$,

$$(e_{000})(a,0) = (e_{000}a, 0) = (a,0) = (ae_{000}, 0) = (a,0)(e_{000}),$$

$$(e_{000})(0,a) = (0, \overline{e_{000}}a) = (0,a) = (0, ae_{000}) = (0,a)(0,e_{000}).$$

2. For each $e_k \in B_{\mathbb{S}}$, $e_k^2 = -e_{0000}$ since for every $e_i \in B_{\mathbb{K}}$

$$(e_i,0)(e_i,0) = (e_ie_i,0) = (-e_{000},0) \text{ and } (0,e_i)(0,e_i) = -(e_i\overline{e_i},0) = (-e_{000},0).$$

3. For distinct non-identity elements $e_k, e_m \in B_{\mathbb{S}}$, $e_ke_m = -e_me_k$.

   (a) For all $e_r, e_s \neq e_{000}$,

   - $(e_r,0)(e_s,0) = (e_re_s,0) = (-e_re_s,0) = -(e_s,0)(e_r,0)$,
   - By Equation (3.1), $(e_r,0)(0,e_s) = (0,\overline{e_r}e_s) = (0,-e_re_s) = -(0,e_s)(e_r,0)$,
   - $(0,e_r)(0,e_s) = -(e_s\overline{e_r},0) = (e_se_r,0) = (-e_re_s,0) = (e_r\overline{e_s},0) = -(0,e_s)(0,e_r)$.

   (b) For all $e_r \neq e_{000}$

   - $(0,e_{000})(e_r,0) = (0,e_re_{000}) = (0,-\overline{e_r}e_{000}) = -(e_r,0)(0,e_{000})$;
   - $(0,e_{000})(0,e_r) = -(e_r\overline{e_{000}},0) = -(e_r,0) = (e_{000}\overline{e_r},0) = -(0,e_r)(0,e_{000})$.

4. For each $e_k \in B_{\mathbb{K}}, k \neq 000$, the set $\{(e_k,0),(0,e_{000}),(0,e_k)\}$ forms a triple system.
   This follows using Property (3) and the fact that

$$(e_k,0)(0,e_k) = (0,\overline{e_k} \cdot e_k) = (0,e_{000}).$$

Proposition 3.4.2 *Let k,m,n be distinct elements in $I_{15} = \{1,2,\ldots,15\}$ with $e_ke_m = e_n$. Then $\langle e_k,e_m,e_n\rangle$ is isomorphic to $\langle i,j,k\rangle$ with $e_k \mapsto i$, $e_m \mapsto j$, $e_n \mapsto k$.*

Proof.

By Properties 3.4.1 (3) and (4), $\forall\ k,m \in I_{15}$, $e_me_k = -e_me_k$ and $e_k^2 = -e_0$. It suffices to show that $e_me_n = e_k$ and $e_ne_k = e_m$. We consider three cases by taking $e_{0i} = (e_i,0)$, $e_{1i} = (0,e_i)$ for $e_i \in B_{\mathbb{K}}$.

1. $e_{0k} = (e_k, 0)$, $e_{0m} = (e_m, 0)$ with $k \neq m$. Here,

   - $e_{0k}e_{0m} = (e_k e_m, 0) = e_{0n}$,
   - $e_{0m}e_{0n} = (e_m, 0)(e_k e_m, 0) = (e_m \cdot e_k e_m, 0) = (-e_m e_m \cdot e_k, 0) = (e_k, 0) = e_{0k}$,
   - $e_{0n}e_{0k} = (e_k e_m, 0)(e_k, 0) = (e_k e_m \cdot e_k, 0) = (-e_m \cdot e_k e_k, 0) = (e_m, 0) = e_{0m}$.

2. $e_{0k} = (e_k, 0)$, $e_{1m} = (0, e_m)$. Then,

   - $e_{0k}e_{1m} = (0, \overline{e_k} e_m) = (0, -e_k e_m) = (0, e_m e_k) = e_{1n}$
   - $e_{1m}e_{1n} = (0, e_m)(0, e_m e_k) = -(e_m e_k \overline{e_m}, 0) = (e_k e_m \overline{e_m}, 0) = e_{0k}$.
   - $e_{1n}e_{0k} = (0, e_m e_k)(e_k, 0) = (0, e_k \cdot e_m e_k) = (0, -e_k e_k e_m) = (0, e_m) = e_{1m}$

3. $e_{1k} = (0, e_k), e_{1m} = (0, e_m)$ with $k \neq m \neq 000$. Then,

   - $e_{1k}e_{1m} = -(e_m \overline{e_k}, 0) = (e_m e_k, 0) = e_{0n}$
   - $e_{1m}e_{on} = (0, e_m)(e_m e_k, 0) = (0, e_m e_k \cdot e_m = (0, -e_k e_m e_m) = (0, e_k) = e_{1k}$
   - $e_{on}e_{1k} = (e_m e_k, 0)(0, e_k) = (0, \overline{e_m e_k} \cdot e_k) = (0, e_k e_m \cdot e_k) = (0, -e_m e_k \cdot e_k) = (0, e_m) = e_{1m}$.

4. If e_k or $e_m = e_{1000} = (0, e_{000})$,the result follows directly by Properties 3.4.1 (3).

□

We now use Properties 3.4.1 and Proposition 3.4.2 to give the multiplication table of the basic elements of the sedenions, Table 3.2. The subscripts are in hexadecimal form for consistency with the 16-dimensional sedenion structure.

For each line in the Fano plane, Figure 3.2, we will construct two Fano planes in the sedenion loop. We use the following lemma.

Table 3.2 Sedenion frame multiplication.

$\cdot$	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
1	1	-0	-3	2	-5	4	7	-6	-9	8	B	-A	D	-C	-F	E
2	2	3	-0	-1	-6	-7	4	5	-A	-B	8	9	E	F	-C	-D
3	3	-2	1	-0	-7	6	-5	4	-B	A	-9	8	F	-E	D	-C
4	4	5	6	7	-0	-1	-2	-3	-C	-D	-E	-F	8	9	A	B
5	5	-4	7	-6	1	-0	3	-2	-D	C	-F	E	-9	8	-B	A
6	6	-7	-4	5	2	-3	-0	1	-E	F	C	-D	-A	B	8	-9
7	7	6	-5	-4	3	2	-1	-0	-F	-E	D	C	-B	-A	9	8
8	8	9	A	B	C	D	E	F	-0	-1	-2	-3	-4	-5	-6	-7
9	9	-8	B	-A	D	-C	-F	E	1	-0	3	-2	5	-4	-7	6
A	A	-B	-8	9	E	F	-C	-D	2	-3	-0	1	6	7	-4	-5
B	B	A	-9	-8	F	-E	D	-C	3	2	-1	-0	7	-6	5	-4
C	C	-D	-E	-F	-8	9	A	B	4	-5	-6	-7	-0	1	2	3
D	D	C	-F	E	-9	-8	-B	A	5	4	-7	6	-1	-0	-3	2
E	E	F	C	-D	-A	B	-8	-9	6	7	4	-5	-2	3	-0	-1
F	F	-E	D	C	-B	-A	9	-8	7	-6	5	4	-3	-2	1	-0

Lemma 3.4.3 *Let $e_k e_m = e_n$ be a line, i.e an (i, j, k)-combination in the octonion basic element multiplication, and let $\{l, r, s, t\} = \{1, 2, 3, 4, 5, 6, 7\} - \{k, m, n\}$. Then,*

1. $\{(e_k, 0), (e_m, 0), (e_n, 0), (0, e_0), (0, e_k), (0, e_m), (0, e_n)\}$ *and*

2. $\{(e_k, 0), (e_m, 0), (e_n, 0), (0, e_l), (0, e_r), (0, e_s), (0, e_t)\}$

are Fano planes in the sedenion multiplication.

Proof.

We will show that each of these sets is closed under multiplication and that each can be fit into a Fano plane.

1. For $i, j \in \{k, m, n\}$,

   - $(e_i, 0)(0, e_i) = (0, \overline{e_i} e_i) = (0, e_0)$; $(e_i, 0)(e_j, 0) = (e_i e_j, 0)$
   - $(e_i, 0)(0, e_j) = (0, -e_i e_j)$.

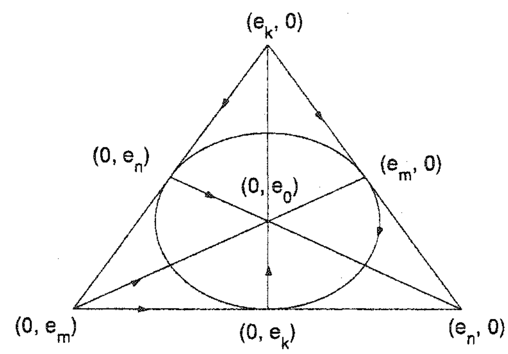

Figure 3.3 Quaternion triplet extesion

The multiplication is represented by the fano plane in Figure 3.3.

2. $\{k, m, n, l, r, s, t\}$ is simply a rearrangement of the set $\{1, 2, 3, 4, 5, 6, 7\}$, so we can assume without loss of generality that the multiplication is represented by the Fano plane in Figure 3.4.

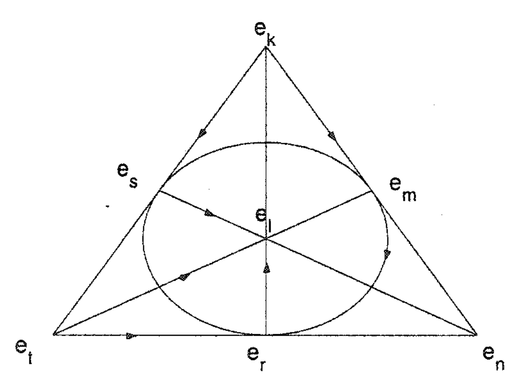

Figure 3.4 Fano representing multiplication in $\{k, m, n, l, r, s, t\}$

For $i, j \in \{k, m, n\}$; $\lambda, \kappa \in \{l, r, s, t\}$;

$(e_i, 0)(e_j, 0) = (e_i e_j, 0)$,

$(e_i, 0)(0, e_\lambda) = (0, \overline{e_i} e_\lambda) = (0, e_\tau)$, for some $\tau \in \{l, r, s, t\}$,

$(0, e_\lambda)(0, e_\kappa) = (e_\kappa e_\lambda, 0) = (e_\tau, 0)$, for some $\tau \in \{k, m, n\}$.

This fits in the Fano plane in Figure 3.5

$\square$

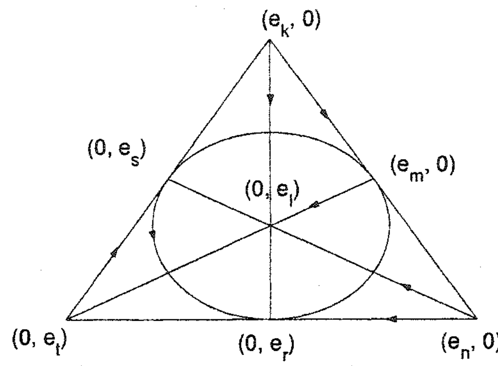

Figure 3.5 Quaternion triplet extesion

Note: The Fano for Quaternion triplet extension 2, Figure 3.5 has all except the arrows for $(e_k, 0)(e_m, 0) = (e_n, 0)$ reversed when compared to the Fano for the octonion frame in Figure 3.2. The octonions of this form are called *quasi* or *pseudo* octonions in [4].

Example 3.4.4 Consider the line $e_{001}e_{011} = e_{010}$ in the octonion multiplication. This gives rise to the Fano planes

1. $\{e_{0001}, e_{0011}, e_{0010}, e_{1000}, e_{1001}, e_{1011}, e_{1010}\}$

2. $\{e_{0001}, e_{0011}, e_{0010}, e_{1100}, e_{1101}, e_{1110}, e_{1111}\}$

in the sedenion multiplication. The multiplication is represented by

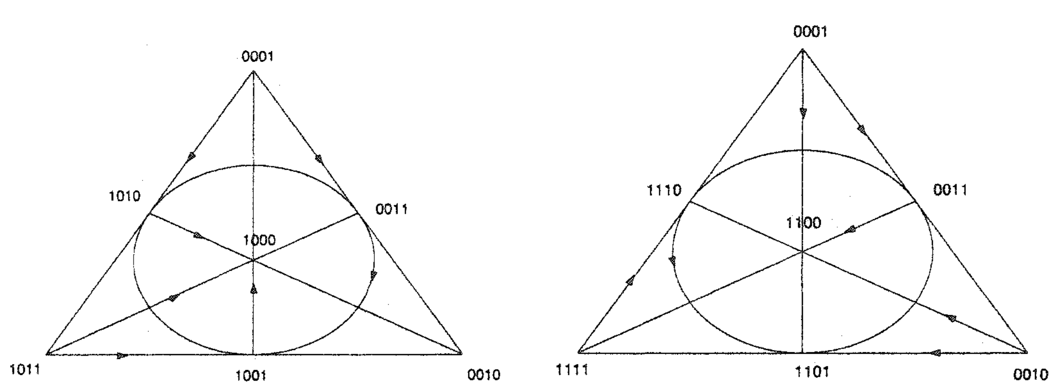

Figure 3.6 The extensions in Sedenions from the triple $001 \to 011 \to 010$

From Lemma 3.4.3, each line in the octonion basic element multiplication gives rise to two Fano planes in the sedenion basic elements multiplication. The set

$$\{(e_1,0),(e_2,0),(e_3,0),(e_4,0),(e_5,0),(e_6,0),(e_7,0)\}$$

also forms a Fano plane in the sedenions. We get a total of $7(2)+1 = 15 = 2(2^3-1)+1 = 2^4-1$ Fano planes. This corresponds to 15 $PG(2,2)$ projective planes in the projective space $PG(3,2)$. Each line is of length 3; it represents an (i,j,k)-combination, and each appears in exactly three Fano planes. Each of the 15 points appears in seven Fano planes and is connected to seven different lines. There is a total of $(7\cdot 15)/3 = 35$ lines. This forms a $PG(3,2)$ projective geometry.

The following lemma gives the characterization of the basic element multiplications in a projective geometry for the 2^n-ons.

Lemma 3.4.5 *Let $k \in \{1,2,\ldots n-2\}$ and suppose the set $A = \{a_1, a_2, \ldots, a_m\}$ forms a $PG(k,2)$ geometry inside a $PG(k+1,2)$ geometry in the 2^n-on basic element multiplication. Let $A \cup B = \{a_1, a_2, \ldots, a_m, b_1, b_2, \ldots, b_{m+1}\}$ be the $PG(k+1,2)$ geometry . Then $\forall\ a_i, a_j \in A,\ b_r, b_s \in B$*

(i) $a_ia_j \in A$

(ii) $a_ib_r \in B$

(iii) $b_rb_s \in A$

Proof.

Since a $PG(k,2)$ geometry is closed under multiplication,

i) Follows immediately from closure of A as a $PG(k,2)$ geometry.

ii) Suppose $a_ib_r = a_k$ for some $a_k \in A$. The set $\{a_i, b_r, a_k\}$ forms a triple system which is associative and so $a_ia_k = a_i(a_ib_r) = (a_ia_i)b_r = -b_r \in B$. A contradiction to the closure of A as a $PG(k,2)$ geometry, so $a_ib_r \in B$.

iii) Suppose on the contrary $b_r b_s = b_t$ for some $b_r,\ b_s,\ b_t \in B$. From (*ii*) above,

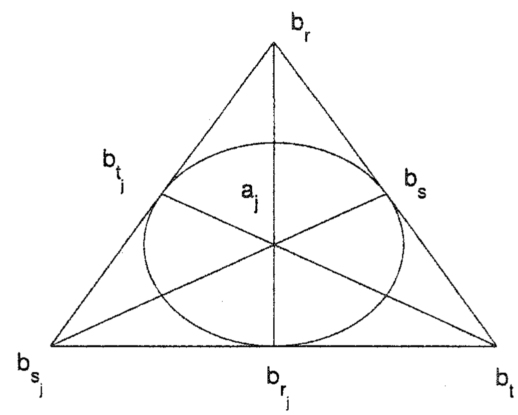

Figure 3.7 Multiplication corresponding to a_j

$\forall\ j \in \{1, 2,\ \ldots,\ m\},\ k \in \{r, s, t\},\ b_k a_j = b_{k_j} \in B$ and the multiplication fits in the Fano plane in Figure 3.7.

Thus, for each $a_j \in A,\ j \in \{1, 2,\ \ldots,\ m\}$, we have 3 distinct elements $b_{r_j},\ b_{s_j},\ b_{t_j} \in B$ as seen in Figure 3.7. We get a total of $3m + 3$ elements in B to fit in all the Fano planes. Since $|B| = m + 1 < 3m + 3$, there is at least one b_{k_j} that appears in at least two Fano planes. Without loss of generality, assume this happens in the Fano planes for a_k, a_m with $k \neq m$ and that $b_{r_k} = b_{s_m} = b_i \notin \{b_r, b_s, b_t\}$. The Fano planes for a_k, a_m are given by Figure 3.8. Now, $b_r a_k = b_i = b_t a_m$ and so

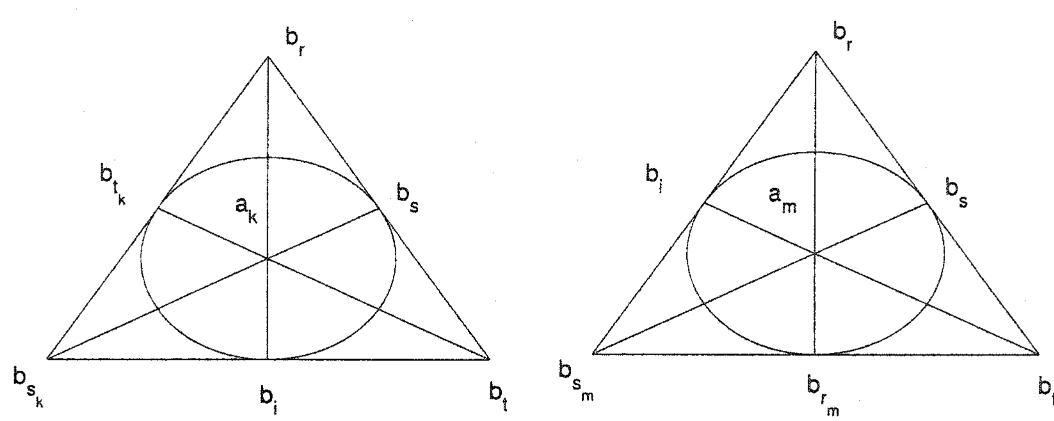

Figure 3.8 Fanos arising from a_k and a_m

$b_{s_k} = b_t b_i = b_t(b_t a_m) = (b_t b_t)a_m = -a_m =$. But $b_{s_k} a_k = b_s$ which implies that $-a_m a_k = b_s$. This contradicts the closure of A as a $PG(k,2)$ geometry.

□

3.5 Nim addition and Nim multiplication

Consider the non-negative integers $\mathbb{Z}^+ = \{0,1,2,3,\ldots\}$. Nim addition and Nim multiplication give a way of defining addition and multiplication in $\mathbb{Z}^+$ to make it a field of characteristic 2. Although we will only need Nim addition in what follows, we describe the Nim multiplication as well for completeness. The symbols $+_N$ and $\odot_N$ will be used for Nim addition and multiplication respectively. The addition table is filled with the condition that before getting $\alpha +_N \beta$, we first must have filled all entries $\alpha' +_N \beta, \quad \alpha +_N \beta'$ for $\alpha' < \alpha,\ \beta' < \beta$. Then $\alpha +_N \beta$ is taken as the least possible number which is consistent with the results being part of the addition table of a field. This kind of addition is used in the game of Nim, and we call it *Nim addition*. The rules of Nim addition simplify to the form:

1. The Nim-sum of a number of distinct 2-powers is their ordinary sum. Thus

$$16 +_N 8 +_N 4 +_N 1 = 29,\ 32 +_N 4 +_N 2 = 38.$$

2. The Nim-sum of two equal numbers is 0.

Table 3.3 gives the Nim addition for the elements from 0 to 15. This table is similar to the multiplication table of the sedenion frame Table 3.2 except for the signs.

Example 3.5.1

1. $21 +_N 7 = (16+4+1) +_N (4+2+1) = 16 +_N 2 = 18.$

2. $189 +_N 165 = (128+32+16+8+4+1) +_N (128+32+4+1) = 16 +_N 8 = 24.$

Table 3.3 Nim addition.

$+_N$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	1	0	3	2	5	4	7	6	9	8	11	10	13	12	15	14
2	2	3	0	1	6	7	4	5	10	11	8	9	14	15	12	13
3	3	2	1	0	7	6	5	4	11	10	9	8	15	14	13	12
4	4	5	6	7	0	1	2	3	12	13	14	15	8	9	10	11
5	5	4	7	6	1	0	3	2	13	12	15	14	9	8	11	10
6	6	7	4	5	2	3	0	1	14	15	12	13	10	11	8	9
7	7	6	5	4	3	2	1	0	15	14	13	12	11	10	9	8
8	8	9	10	11	12	13	14	15	0	1	2	3	4	5	6	7
9	9	8	11	10	13	12	15	14	1	0	3	2	5	4	7	6
10	10	11	8	9	14	15	12	13	2	3	0	1	6	7	4	5
11	11	10	9	8	15	14	13	12	3	2	1	0	7	6	5	4
12	12	13	14	15	8	9	10	11	4	5	6	7	0	1	2	3
13	13	12	15	14	9	8	11	10	5	4	7	6	1	0	3	2
14	14	15	12	13	10	11	8	9	6	7	4	5	2	3	0	1
15	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

Nim addition can be done by first writing the numbers in binary form and then "*adding without carrying over.*"

Example 3.5.2

1. $21 +_N 7 = 10101 +_N 111 = 10010 = 18$

2. $189 +_N 165 = 10111101 +_N 10100101 = 00011000 = 24.$

Nim addition is used in the Nim multiplication, just as in ordinary fields. Similar rules are set for the multiplication as in the addition. A *Fermat* 2-power is a number of the form $2^{2n} = 4^n$. The rules for the Nim multiplication are:

1. The Nim-product of a number of distinct Fermat 2-powers is their ordinary product, i.e $2^{2n_1} \cdot 2^{2n_2} \ \cdots \ 2^{2n_k} = 2^{2(n_1+n_2+\cdots+n_k)}$

2. The square of a Fermat 2-power is its *sesquimultiple.* The sesquimultiple of k is $3k/2$, and so $2^{2n} \cdot 2^{2n} = 3((2^{2n})/2 = 3(2^{2n-1})$.

The Nim-product is done by first writing each number as a sum of the Fermat 2-powers $1, 4, 16, 64, \ldots$, and then multiplying using the rules given above. It will be noted here that each number in terms of Fermat 2-powers takes the form

$$a_n \cdot 2^{2n} + a_{n-1} \cdot 2^{2(n-1)} + \cdots + a_1 \cdot 2^2 + a_0 \quad \text{where } a_i \in \{0, 1, 2, 3\}. \tag{3.2}$$

Table 3.4 gives the Nim product of the integers 0 to 15. Note that given the multiplication table of 0–4, one can fill in the table from 0–8 and then 0–16 and the process continues.

Table 3.4 Nim multiplication.

$\odot_N$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
2	0	2	3	1	8	10	11	9	12	14	15	13	4	6	7	5
3	0	3	1	2	12	15	13	14	4	7	5	6	8	11	9	10
4	0	4	8	12	6	2	14	10	11	15	3	7	13	9	5	1
5	0	5	10	15	2	7	8	13	3	6	9	12	1	4	11	14
6	0	6	11	13	14	8	5	3	7	1	12	10	9	15	2	4
7	0	7	9	14	10	13	3	4	15	8	6	1	5	2	12	11
8	0	8	12	4	11	3	7	15	13	5	1	9	6	14	10	2
9	0	9	14	7	15	6	1	8	5	12	11	2	10	3	4	13
10	0	10	15	5	3	9	12	6	1	11	14	4	2	8	13	7
11	0	11	13	6	7	12	10	1	9	2	4	15	14	5	3	8
12	0	12	4	8	13	1	9	5	6	10	2	14	11	7	15	3
13	0	13	6	11	9	4	15	2	14	3	8	5	7	10	1	12
14	0	14	7	9	5	11	2	12	10	4	13	3	15	1	8	6
15	0	15	5	10	1	14	4	11	2	13	7	8	3	12	6	9

Example 3.5.3 Suppose we have the multiplication table for 0–8 filled and we need $15 \odot_N 12$, Then

$15 \odot_N 12 = (8 +_N 4 +_N 2 +_N 1) \odot (8 +_N 4) = (8 \odot_N 8) +_N (8 \odot_N 4) +_N (4 \odot_N 8) +_N (4 \odot_N 4) +_N (2 \odot_N 8) +_N (2 \odot_N 4) +_N (1 \odot_N 8) +_N (1 \odot_N 4) +_N = 13 +_N 6 +_N 12 +_N 8 +_N 8 +_N 4 = (8 +_N 4 +_N 1) +_N (4 +_N 2) +_N (8 +_N 4) +_N 4 = 3$

Properties 3.5.4

i) $\alpha +_N \beta = \alpha +_N \gamma \quad \text{iff } \beta = \gamma$

ii) $\alpha +_N \beta = \beta +_N \alpha$

iii) $(\alpha +_N \beta) +_N \gamma = \alpha +_N (\beta +_N \gamma)$

iv) $\alpha +_N \alpha = 0, \quad -\alpha = \alpha$

v) $\alpha \odot_N 0 = 0, \quad \alpha \odot_N 1 = \alpha, \quad \alpha \odot_N \beta = \beta \odot_N \alpha$

vi) $(\alpha +_N \beta) \odot_N \gamma = \alpha \odot_N \gamma + \beta \odot_N \gamma$

vii) $(\alpha \odot_N \beta) \odot_N \gamma = \alpha \odot_N (\beta \odot_N \gamma)$

We use Nim addition to describe basic multiplications in the 2^n-ons.

Lemma 3.5.5 *Let $\{e_0,\ e_1,\ \ldots,\ e_{2^n-1}\}$ be the basic elements of the 2^n-ons. Then $e_k,\ e_m,\ e_{k+_N m}$ forms a triple system, i.e $e_k \cdot e_m = \pm e_{k+_N m}$.*

Proof.

The result holds in the complex, quaternion and Cayley numbers as seen in Table 3.3. We assume that the result holds in the 2^{n-1}-ons with basic elements $\{e'_0,\ e'_1,\ \ldots,\ e'_{2^{n-1}-1}\}$. The basic elements of the 2^n-ons are

$\{e_0 = (e'_0, 0),\ e_1 = (e'_1, 0),\ \ldots,\ e_i = (e'_i, 0),\ \ldots,\ e_{2^{n-1}-1} = (e'_{2^{n-1}-1}, 0),$
$e_{2^{n-1}} = (0, e'_0),\ e_{2^{n-1}+1} = (0, e'_1),\ \ldots,\ e_{2^{n-1}+i} = (0, e_i),\ \ldots,\ e_{2^n-1} = (0, e'_{2^{n-1}-1})\}.$

We consider three cases for $e_k \cdot e_m$ in the 2^n-ons:

1. $e_k = (e'_k, 0),\ e_m = (e'_m, 0)$. Then,

$$e_k \cdot e_m = (e'_k, 0) \cdot (e'_m, 0) = (e'_k \cdot e'_m, 0) = (e'_{k+_N m}, 0) = e_{k+_N m}.$$

2. $e_k = (e'_k, 0),\ e_m = (0, e'_l),$ here $m = 2^{n-1} + l$. Then,

$$e_k e_m = (e'_k, 0)(0, e'_l) = (0, e'_k e'_l) = (0, e'_{k+_N l}) = e_{2^{n-1}+(k+_N l)} = e_{k+_N m}.$$

Also $e_m e_k = (0, e'_l)(e'_k, 0) = (0, e'_k e'_l) = (0, e'_{k+_N l}) = e_{2^{n-1}+l\ +_N k} = e_{m+_N k}$.

3. $e_k = (0, e'_r),\ e_m = (0, e'_l)$, here $k = 2^{n-1} + r,\ m = 2^{n-1} + l$. Then,

$$e_k e_m = (0, e'_r)(0, e'_l) = (-e'_l \overline{e'_r}, 0) = \pm(e'_l e'_r, 0) = \pm(e_{l+_N r}, 0) = e_{k+_N m},$$

since $k +_N m = (2^{n-1} + r) + (2^{n-1} + l) = r +_N l$.

We conclude that the set $\{e_k, e_m, e_{k+_N m}\}$ forms a triple system in the 2^n-ons. □

3.6 Main Theorem

Theorem 3.6.1 *The multiplication of the non-identity signed basisc elements in the 2^n-ons can be represented by the projective geometry $PG(n-1, 2)$.*

Proof.

The basic element multiplications in the complex, quaternions and Cayley numbers fit in their respective projective geometries, namely a point, a line with three points, and a Fano plane respectively. This sets up the basis for an induction proof. We assume that the multiplication in the 2^{n-1}-ons fits in the $PG(n-2, 2)$ projective geometry, and show the result holds in the 2^n-ons.

By the induction hypothesis, we assume that for $k = 1, 2, \ldots, n-3$, a $PG(k, 2)$ projective geometry in the 2^{n-2}-ons leads to a $PG(k+1, 2)$ geometry in the 2^{n-1}-ons. It suffices to show that the multiplication in the 2^n-ons fits in $2^n - 1$, $PG(n-2, 2)$ projective geometries.

Let $E = \{e_0,\ e_1,\ \ldots,\ e_{2^{n-1}-1}\}$ be the basic elements of the 2^{n-1}-ons, which by assumption fit in the $PG(n-2, 2)$ projective geometry comprising of $2^{n-1}-1$ $PG(n-3, 2)$

projective geometries. We will show that each of these $PG(n-3,2)$ projective geometries gives rise to two distinct $PG(n-2,2)$. This will give a total of $2(2^{n-1}-1) = 2^n - 2$, $PG(n-2,2)$ projective geometries. These together with the $PG(n-2,2)$ geometry $\{(e_0,0),\ (e_1,0),\ \ldots,\ (e_{2^{n-1}-1},0)\}$ give 2^n-1, $PG(n-2,2)$ projective geometries which make up the $PG(n-1,2)$ projective geometry in the 2^n-ons.

Let $A = \{a_1,\ a_2,\ \ldots,\ a_{2^{n-2}-1}\}$ be a $PG(n-3,2)$ in E. We construct two $PG(n-2,2)$ projective geometries from A.

1. The elements

$$(a_1,0),\ (a_2,0),\ \ldots,\ (a_{2^{n-2}-1},0),\ (0,e_0),\ (0,a_1),\ (0,a_2),\ \ldots,\ (0,a_{2^{n-2}-1})$$

form a $PG(n-2,2)$ by the induction hypothesis. Note that the multiplication in A corresponds to the multiplication in the non-identity basic elements $\{e_1,\ e_2,\ \ldots,\ e_{2^{n-2}-1}\}$ of the 2^{n-2}-ons as a $PG(n-3,2)$ projective geometry. Now by the induction hypothesis

$$\{(e_1,0),\ (e_2,0),\ \ldots,\ (e_{2^{n-2}-1},0),\ (0,e_0),\ (0,e_1),\ (0,e_2),\ \ldots,\ (0,e_{2^{n-2}-1})\}$$

forms a $PG(n-2,2)$ geometry in the 2^{n-1}-ons, and hence in the same way

$$\{(a_1,0),\ (a_2,0),\ \ldots,\ (a_{2^{n-2}-1},0),\ (0,a_1),\ (0,a_2),\ \ldots,\ (0,a_{2^{n-2}-1})\}$$

forms a $PG(n-2,2)$.

2. Let $B = \{b_1,\ b_2,\ \ldots,\ b_{2^{n-2}}\} = E - A$. Then,

$$M = \{(a_1,0),\ (a_2,0),\ \ldots,\ (a_{2^{n-2}-1},0),\ (0,b_1),\ (0,b_2),\ \ldots,\ (0,b_{2^{n-2}})\}$$

forms a $PG(n-2,2)$.

By Lemma 3.4.5 and Lemma 3.5.5,

$$(a_i,0)(a_j,0),\ (0,b_r)(0,b_t) \in \{(a_1,0),\ (a_2,0),\ \ldots,\ (a_{2^{n-2}-1},0)\}$$

and

$$(a_i, 0)(0, b_r) \in \{(0, b_1),\ (0, b_2),\ \ldots,\ (0, b_{2^{n-2}})\}.$$

Without taking the sign into consideration, the multiplication in M corresponds to that in E. Thus M forms a $PG(n-2, 2)$ geometry.

Each $PG(n-3, 2)$ geometry as in $(A, 0) = \{(a_1, 0),\ (a_2, 0)\ \ldots,\ (a_{2^{n-2}-1}, 0)\}$ is contained in three $PG(n-2, 2)$ geometries $(E, 0)$, $(A, 0) \cup \{(0, e_0)\} \cup (0, A)$ and $(A, 0) \cup (0, B)$. We have a total of $2^n - 1$, $PG(n-2, 2)$ geometries with each $PG(n-3, 2)$ geometry contained in three $PG(n-2, 2)$ geometries. This forms the desired $PG(n-1, 2)$ projective geometry and the proof is complete.

$\square$

3.7 Gaussian Numbers and q-analogues

3.7.1 Introduction

Let $V_n(q)$ denote an n-dimensional vector space over the field $GF(q)$ of q elements. The number of maximal chains (chain of size $n+1$ containing one subspace of each possible dimension) of subspaces in $V_n(q)$ is given by

$$M(n, q) = \frac{(q^n - 1)(q^{n-1} - 1) \cdots (q^2 - 1)(q - 1)}{(q - 1)^n}.$$

$M(n, q)$ is a polynomial in q for each integer n. When $q = p^r$, p prime we have the number of maximal chains in the poset $PG(n, q)$. When $q = 1$, $M(n, 1) = n!$ which is the number of maximal chains in the poset of an n-set. The *Gaussian number* $\begin{bmatrix} n \\ k \end{bmatrix}_q$ can be defined as the number of k dimensional subspaces of $V_n(q)$. It is also called the

Gaussian coefficient or *polynomial*, and is given by

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{M(n,q)}{M(k,q)M(n-k,q)} = \frac{(q^n-1)(q^{n-1}-1)\cdots(q^{n-k+1}-1)}{(q^k-1)(q^{k-1}-1)\cdots(q^2-1)(q-1)}.$$

Clearly $\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ n-k \end{bmatrix}_q$, therefore the number of subspaces of order q^k is equal to the number of subspaces of order q^{n-k} for $0 \leq k \leq n$. Also, $\begin{bmatrix} n \\ k \end{bmatrix}_{q=1} = \begin{pmatrix} n \\ k \end{pmatrix}$.

Example 3.7.1 $\begin{bmatrix} 6 \\ 3 \end{bmatrix}_q = q^9 + q^8 + 2q^7 + 3q^6 + 3q^5 + 3q^4 + 3q^3 + 2q^2 + q + 1$ *and* $\begin{bmatrix} 6 \\ 3 \end{bmatrix}_1 = 1+1+2+3+3+3+3+2+1+1 = 20 = \begin{pmatrix} 6 \\ 3 \end{pmatrix}$.

A much more detailed look on Gaussian polynomials can be found in [18, Chap.24].

Corollary 3.7.2 *Suppose* $A = \{a_1, a_2, \ldots, a_{2^k-1}\}$ *forms a* $PG(k-1,2)$ *geometry in the* $PG(n-2,2)$ *geometry under the* 2^{n-1}*-on multiplication, and suppose this geometry appears in* m *of the* $PG(n-3,2)$ *geometries in* $PG(n-2,2)$*. Then the* $PG(k-1,2)$ *geometry* $\{(a_1,0),(a_2,0),\ldots,(a_{2^k-1},0)\}$ *in the* 2^n*-on multiplication appears in* $2m+1$*,* $PG(n-2,2)$ *geometries.*

Proof.

Let $E = \{e_0, e_1, e_2, \ldots, e_{2^{n-1}-1}\}$ be the basic elements of the 2^{n-1}-ons and let $B = \{a_1, a_2, \ldots, a_s, a_{s+1}, \ldots, a_{2^{n-2}-1}\}$ be a $PG(n-3,2)$ geometry containing the $PG(k-1,2)$ geometry A in the 2^{n-1}-on multiplication. There are two $PG(n-2,2)$ geometries in the 2^n-on multiplication arising from this $PG(n-3,2)$ geometry:

1) $\{(a_1,0),(a_2,0),\ldots,(a_t,0),(0,1),(0,a_1),(0,a_2),\ldots,(0,a_{2^{n-2}-1})\}$

2) Let $C = E - B = \{c_1, c_2, \ldots, c_{2^{n-2}}\}$ then $(a_1, 0), (a_2, 0), \ldots, (a_{2^{n-2}-1}, 0), (0, c_1), (0, c_2), \ldots, (0, c_{2^{n-2}})\}$ forms a $PG(n-12, 2)$ geometry. Clearly the $PG(k-1, 2)$, $\{(a_1, 0), (a_2, 0), \ldots, (a_{2^k-1}, 0)\}$ is contained in each of these $PG(n-2, 2)$ geometries. There are $2m$, $PG(n-2, 2)$ geometries containing $\{(a_1, 0), (a_2, 0), \ldots, (a_{2^k-1}, 0)\}$. These together with the $PG(n-2, 2)$ geometry $\{(e_1, 0), (e_2, 0), \ldots, (e_{2^{n-1}-1}, 0)\}$, comprise $2m+1$, $PG(n-2, 2)$ geometries. □

Corollary 3.7.3 *For $2 \leq k \leq n$, each $P(n-k, 2)$ geometry appears in $2^k - 1$, $P(n-2, 2)$ geometries in the 2^n-on frame multiplication. Thus, each subloop of order 2^{n-k} is contained in $2^k - 1$ maximal subloops in the 2^n-on frame multiplication.*

Proof.

The maximal subloops in the 2^{n-k+1}-on frame are of order 2^{n-k}, i.e the $P(n-k-1, 2)$ geometry is maximal in the $P(n-k, 2)$ geometry. The $P(n-k-1, 2)$ geometry arising from this geometry appears in $3 = 2(1) + 1 = 2^2 - 1$ geometries in the 2^{n-k+2}-ons by Corollary 3.7.2. In the 2^{n-k+3}-ons, this geometry appears in $2(2^2 - 1) + 1 = 2^3 - 1$ $P(n-k+1, 2)$ geometries. Repeating this process, and applying Corollary 3.7.2 each time, the $P(n-k-1, 2)$ geometry appears in $2^k - 1$, $P(n-2, 2)$ geometries in the 2^n-ons.

□

Corollary 3.7.4 *The number of subloops of order 2^{n-k} in the 2^n-on frame multiplication is*

$$\begin{bmatrix} n \\ n-k \end{bmatrix}_2 = \frac{M(n,2)}{M(n-k,2)M(k,2)} = \frac{(2^n-1)(2^{n-1}-1)\cdots(2^{n-k+1}-1)}{(2^k-1)(2^{k-1}-1)\cdots(2^2-1)(2-1)}.$$

Proof.

Counting the number of subloops of order 2^k is equivalent to counting the number of $PG(k, 2)$ geometries in the $PG(n-1, 2)$ geometry formed by the 2^n-on basic element multiplication. There are $2^n - 1$ maximal, $PG(n-2, 2)$ geometries which correspond to

$2^n - 1$ subloops of order 2^{n-1}. There are $2^{n-1} - 1$, $PG(n-3, 2)$ geometries inside each $PG(n-2, 2)$ geometry and by Corollary 3.7.3 each appears in $3 = 2^2 - 1$, $PG(n-2, 2)$ geometries. This gives a total of

$$(2^n - 1)\frac{(2^{n-1} - 1)}{3} = \frac{(2^n - 1)(2^{n-1} - 1)}{(2^1 - 1)(2^2 - 1)} = \begin{bmatrix} n \\ n-2 \end{bmatrix}_2 .$$

.

Suppose there are $\begin{bmatrix} n \\ k+1 \end{bmatrix}_2$ subloops of order 2^{n-k+1}, i.e. $PG(n-k, 2)$ geometries in the $PG(n-1, 2)$ geometry. Each $PG(n-k, 2)$ geometry has $2^{n-k+1} - 1$, $PG(n-k-1, 2)$ geometries and each appears in $2^k - 1$ of the $PG(n-2, 2)$ geometries. Therefore, the total number of $PG(n-k, 2)$ geometries, i.e. subloops of order 2^{n-k}, is

$$\frac{(2^{n-k+1} - 1)}{2^k - 1} \cdot \begin{bmatrix} n \\ n-2 \end{bmatrix}_2 = \begin{bmatrix} n \\ n-k \end{bmatrix}_2 .$$

□

3.7.2 Counting subloops of order 2^k in the 2^n-on basic multiplication

Tables 3.5 and Table 3.6 give the number of subloops in the 2^n-on frame for $n \geq 0$:

Table 3.5 Number of subloops of order 2^k in the 2^n-ons

$n \backslash k$	1	2	3	4	5	6	7	8	9	Total
0	1									1
1	1	1								2
2	1	3	1							5
3	1	7	7	1						16
4	1	15	35	15	1					67
5	1	31	155	155	31	1				374
6	1	63	651	1395	651	63	1			2825
7	1	127	2667	11811	11811	2667	127	1		29212
8	1	255	10795	97155	200787	97155	10795	255	1	427482

Table 3.6 Pascal triangle for number of subloops

2^0									1								
2^1								1		1							
2^2							1		3		1						
2^3						1		7		7		1					
2^4					1		15		35		15		1				
2^5				1		31		155		155		31		1			
2^6			1		63		651		1395		651		63		1		
2^7		1		127		2667		11811		11811		2667		127		1	
2^8	1		255		10795		97155		200787		97155		10795		255		1

CHAPTER 4. SIGN MATRICES FOR FRAMES OF 2^n-ONS

In the present chapter, we will investigate the sign matrices for the frame multiplication in the 2^n-ons. We will consider the Smith-Conway (1.17), (1.18) and Cayley-Dickson (4.9) multiplications. We show that the sign matrices in these two multiplications are skew Hadamard. Kronecker products are also introduced, and it is shown that the sign matrices for the quaternions and octonions are equivalent to Kronecker products.

4.1 Introduction

The importance of orthogonal matrices in modern discrete mathematics and its applications is well known for many applied problems. Hadamard matrices were originally investigated in this category of matrices by Sylvester in 1867. The original investigations were connected with the linear algebra problem of maximizing determinants. Jacques Hadamard (1893) proved that if A is an $n \times n$ matrix, then

$$|detA|^2 \leq \prod_{i=1}^{n} \sum_{j=1}^{n} |A_{ij}|^2 . \tag{4.1}$$

He discovered that if $|A_{ij}| \leq 1$ and equality holds in (4.1), then A is (± 1)-valued and $n = 1, 2$ or $4m$. Matrices satisfying the equality in (4.1) came to be called Hadamard matrices. It was later realized that Hadamard matrices were very helpful in problems relating to signal processing, machine learning, and information theory. There is an interrelation between Hadamard matrices and combinatorial configurations such as block designs, Latin squares, error-correcting codes and finite geometry [12].

With the wide application of Hadamard matrices, mathematicians have been devoted to finding new methods of construction and their underlying properties. Throughout this chapter, H' will denote the transpose of the matrix H.

Definition 4.1.1 A *Hadamard* matrix of order n is an $n \times n$ matrix H with entries ± 1 such that $HH' = nI$. A Hadamard matrix is *normalized* if the first row and first column consist entirely of $+$ 1's.

The following are some fundamental properties of Hadamard matrices:

Properties 4.1.2 *Let H be an $n \times n$ Hadamard matrix.*

1. *Any two columns of H are orthogonal of weight n.*

2. *If some rows or columns of H are permuted, the resulting matrix is Hadamard.*

3. *If some rows or columns are multiplied by -1, the resulting matrix is still Hadamard.*

4. *If H is Hadamard, H' is Hadamard.*

Definition 4.1.3 Let H_1 and H_2 be Hadamard matrices. If H_2 can be obtained from H_1 using Properties 4.1.2 (2), (3) and (4), then H_1 and H_2 are said to be *equivalent*. By using these properties, every Hadamard matrix can be transformed to an Hadamard matrix such that its first row (resp. column) is all 1 which is called row-normal (resp. column-normal). An Hadamard matrix is called *normal* if it is both row and column normal.

4.2 Construction of Hadamard matrices

The first work on Hadamard matrices was by Sylvester, who in 1867, proposed an iterative method for constructing Hadamard matrices of order 2^k. Scarpis (1898) proved that if $p \equiv 3 \pmod 4$ or $p \equiv 1 \pmod 4$ is a prime, then there is a Hadamard matrix of order $p+1,\ p+3$ respectively [1, p. 2].

4.3 Sign matrices under the Smith-Conway Multiplication

In this section, we will show that the sign matrix of the frame multiplication in the 2^n-ons under the Smith-Conway doubling process (3.1) is a Hadamard matrix.

Definition 4.3.1 Let $A = \{a_1, a_2, \ldots, a_n\}$ be a multiplicative quasigroup in an abelian group with $a_i \cdot a_j \in \{a_k,\ -a_k\}$. The *sign matrix* associated with A is the $n \times n$ matrix S with $S_{ij} = 1$ if $a_i \cdot a_j = a_k$ and $S_{ij} = -1$ if $a_i \cdot a_j = -a_k$.

We now consider the sign matrices of the 2^n-ons frame under the Smith-Conway multiplication Equation (3.1)

4.3.1 Complex numbers

The frame for the complex numbers is

$$\pm\{e_0 = 1, e_1 = i\}.$$

The sign matrix for the frame multiplication is the Hadamard matrix:

$$S_{\mathbb{C}} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad \text{and} \quad S_{\mathbb{C}} S'_{\mathbb{C}} = 2I. \tag{4.2}$$

4.3.2 Quaternions

Considering the quaternions (2^2-ons) with the frame

$$\{\pm e_{j_1 j_2} \mid j_k = 0, 1\}$$

where $e_{01} \cdot e_{10} = -e_{11}$, the multiplication is represented by

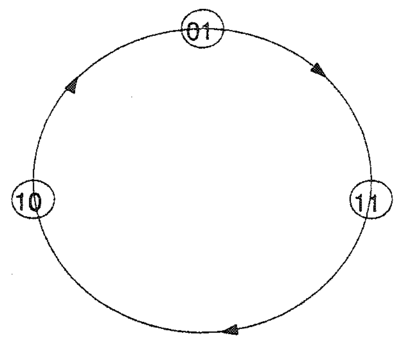

Figure 4.1 Smith- Conway multiplication of quaternion frame

The direction of the arrows show a positive result, while against the arrows show a negative result. For example $e_{10} \cdot e_{11} = -e_{01}$. The sign matrix for the multiplication is given by:

$$S_{\mathbb{H}} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \end{bmatrix} \quad \text{and} \quad S_{\mathbb{H}} S'_{\mathbb{H}} = 4I_4. \tag{4.3}$$

4.3.3 Cayley numbers

The frame of the Cayley numbers (2^3-ons) is

$$B_{\mathbb{K}} = \{e_{000} = (e_{00}, 0),\ e_{001} = (e_{01}, 0),\ e_{010} = (e_{10}, 0),\ e_{011} = (e_{11}, 0),$$
$$e_{100} = (0, e_{00}), e_{101} = (0, e_{01}),\ e_{110} = (0, e_{10}),\ e_{111} = (0, e_{11})\}$$

For $k \neq 000$, we have $e_k^2 = -(e_{00}, 0) = -e_{000}$ and for distinct non-identity elements $e_r \cdot e_k \in \{e_l, \ -e_l\}$, the set $\{e_k, e_r, e_l\}$ forms a triple system i.e a representation as in Figure 4.1. The multiplication is given by the Fano plane, Figure 4.2.

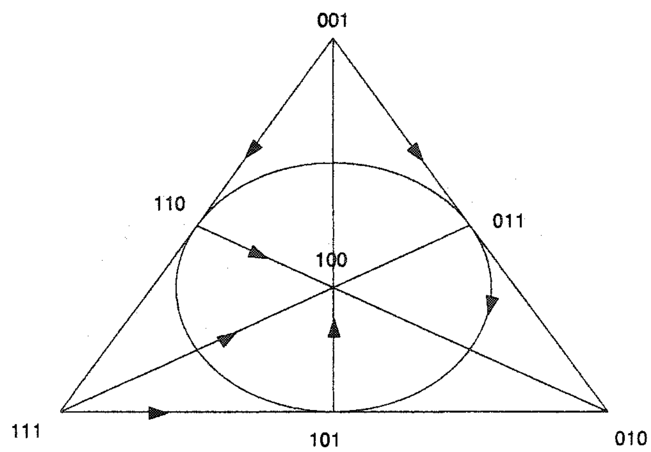

Figure 4.2 Smith-Conway multiplication of octonion frame

The sign matrix for the octonion frame multiplication is given by:

$$S_{\mathbb{K}} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 \\ 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \end{bmatrix} \tag{4.4}$$

and $S_{\mathbb{K}} \cdot S'_{\mathbb{K}} = 8I_8 = 2^3 I_8$.

4.3.4 Sedenions

The construction of the sedenions (2^4-ons) arises from the doubling of the Cayley numbers. From the multiplication table of the sedenion frame Table 3.2, we get the sign matrix:

$$
S_{\mathbb{S}} = \left[\begin{array}{cccccccc|cccccccc}
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & 1 & -1 & -1 & 1 \\
1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & 1 & 1 & -1 & -1 \\
1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & 1 & -1 & 1 & -1 \\
1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 \\
1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 \\
1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 \\
1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 \\
\hline
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \\
1 & -1 & 1 & -1 & 1 & -1 & -1 & 1 & 1 & -1 & 1 & -1 & 1 & -1 & -1 & 1 \\
1 & -1 & -1 & 1 & 1 & 1 & -1 & -1 & 1 & -1 & -1 & 1 & 1 & 1 & -1 & -1 \\
1 & 1 & -1 & -1 & 1 & -1 & 1 & -1 & 1 & 1 & -1 & -1 & 1 & -1 & 1 & -1 \\
1 & -1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 & 1 \\
1 & 1 & -1 & 1 & -1 & -1 & -1 & 1 & 1 & 1 & -1 & 1 & -1 & -1 & -1 & 1 \\
1 & 1 & 1 & -1 & -1 & 1 & -1 & -1 & 1 & 1 & 1 & -1 & -1 & 1 & -1 & -1 \\
1 & -1 & 1 & 1 & -1 & -1 & 1 & -1 & 1 & -1 & 1 & 1 & -1 & -1 & 1 & -1
\end{array}\right] \tag{4.5}
$$

Here $S_{\mathbb{S}} \cdot S'_{\mathbb{S}} = 16 I_{16} = 2^4 I_{16}$.

It is motivating to ask if the sign matrix of the frame multiplication is Hadamard for the general 2^n-ons. The following theorem shows that this is indeed the case.

Theorem 4.3.2 *The sign matrix for the frame in the 2^n-ons under the Smith-Conway multiplication is a $2^n \times 2^n$ Hadamard matrix.*

Proof.

From the preceding discussion, the sign matrix is Hadamard for $n \leq 4$. This sets up the basis for an induction proof. Let S, V be the sign matrices for the frame multiplication in the 2^n-ons and 2^{n+1}-ons respectively. By the induction hypothesis, we assume that S is Hadamard, and show that V is Hadamard. Now,

$$V = \left[\begin{array}{c|c} \text{A} & \text{B} \\ \hline \text{C} & \text{D} \end{array}\right], \tag{4.6}$$

where A, B, C and D are $2^n \times 2^n$ matrices. Let $\{e_0 = 1,\ e_1, \ldots,\ e_{2^n-1}\}$ be the basic elements of the 2^n-ons, then

$$\{(e_0, 0),\ (e_1, 0), \ldots,\ (e_{2^n-1}, 0),\ (0, e_0),\ (0, e_1), \ldots,\ (0, e_{2^n-1}\}$$

are the basic elements of the 2^{n+1}-ons.

$A_{ij} = \text{sign}[(e_{i-1}, 0)(e_{j-1}, 0)] = \text{sign}(e_{i-1} \cdot e_{j-1}, 0) = \text{sign}(e_{i-1} \cdot e_{j-1}) = S_{ij}$ and so $A = S$.

$B_{ij} = \text{sign}[(e_{i-1}, 0)(0, e_{j-1})] = \text{sign}(0, \overline{e_{i-1}} \cdot e_{j-1}) = \text{sign}(\overline{e_{i-1}} \cdot e_{j-1})$. Thus, $B_{1j} = \text{sign}(e_{j-1}) = 1$, and for $i \neq 1$, $B_{ij} = \text{sign}(-e_{i-1} \cdot e_{j-1}) = -S_{ij}$.

$$A = \left[\begin{array}{c|cccc} 1 & 1 & . & . & 1 \\ \hline 1 & & & & \\ . & & & & \\ . & & S_{ij} & & \\ . & & & & \\ 1 & & & & \end{array}\right] = S; \qquad B = \left[\begin{array}{c|cccc} 1 & 1 & . & . & 1 \\ \hline -1 & & & & \\ . & & & & \\ . & & -S_{ij} & & \\ . & & & & \\ -1 & & & & \end{array}\right].$$

$C_{ij} = \text{sign}[(0, e_{i-1})(e_{j-1}, 0)] = \text{sign}(0, e_{j-1}.e_{i-1}) = \text{sign}(e_{j-1}.e_{i-1}) = S_{ji}$ and so $C = S'$.

$D_{ij} = \text{sign}[(0, e_{i-1})(0, e_{j-1})] = \text{sign}(-e_{j-1}.\overline{e_{i-1}}, 0) = \text{sign}(-e_{j-1}.\overline{e_{i-1}})$. Thus, $D_{1j} = \text{sign}(-e_{j-1}) = -1$, for $i \neq 1$, $D_{ij} = \text{sign}(e_{j-1}.e_{i-1}) = S_{ji}$.

$$C = \left[\begin{array}{c|cccc} 1 & 1 & . & . & . & 1 \\ \hline 1 & & & & & \\ . & & & & & \\ . & & & S_{ji} & & \\ . & & & & & \\ 1 & & & & & \end{array}\right] = S'; \qquad D = \left[\begin{array}{c|cccc} -1 & -1 & . & . & . & -1 \\ \hline 1 & & & & & \\ . & & & & & \\ . & & & S_{ji} & & \\ . & & & & & \\ 1 & & & & & \end{array}\right].$$

Combining these results into Equation (4.6) we have

$$V = \left[\begin{array}{cccccc|cccccc} 1 & 1 & . & . & . & 1 & 1 & 1 & . & . & . & 1 \\ 1 & & & & & & -1 & & & & & \\ . & & & & & & . & & & & & \\ . & & & S_{ij} & & & . & & & -S_{ij} & & \\ . & & & & & & . & & & & & \\ 1 & & & & & & -1 & & & & & \\ \hline 1 & 1 & . & . & . & 1 & -1 & -1 & . & . & . & -1 \\ 1 & & & & & & 1 & & & & & \\ . & & & & & & . & & & & & \\ . & & & S_{ji} & & & . & & & S_{ji} & & \\ . & & & & & & . & & & & & \\ 1 & & & & & & 1 & & & & & \end{array}\right].$$

We perform a row permutation $R_1 \leftrightarrow R_{2^n+1}$, i.e $R_{(e_0,0)} \leftrightarrow R_{(0,e_0)}$, to get an equivalent matrix:

$$H = \left[\begin{array}{ccccc|ccccc}
1 & 1 & \cdot & \cdot & \cdot\ 1 & -1 & -1 & \cdot & \cdot & \cdot\ -1 \\
\cdot & & & & & \cdot & & & & \\
\cdot & & S_{ij} & & & \cdot & & -S_{ij} & & \\
\cdot & & & & & \cdot & & & & \\
1 & & & & & -1 & & & & \\
\hline
1 & 1 & \cdot & \cdot & \cdot\ 1 & 1 & 1 & \cdot & \cdot & \cdot\ 1 \\
\cdot & & & & & \cdot & & & & \\
\cdot & & S_{ji} & & & \cdot & & S_{ji} & & \\
\cdot & & & & & \cdot & & & & \\
1 & & & & & 1 & & & &
\end{array}\right] = \left[\begin{array}{c|c} \mathrm{S} & \text{-S} \\ \hline S' & S' \end{array}\right]. \tag{4.7}$$

$$\text{Now } HH' = \left[\begin{array}{c|c} \mathrm{S} & \text{-S} \\ \hline S' & S' \end{array}\right] \left[\begin{array}{c|c} S' & \mathrm{S} \\ \hline -S' & \mathrm{S} \end{array}\right] = \left[\begin{array}{c|c} SS' + SS' & S^2 - S^2 \\ \hline (S')^2 - (S')^2 & S'S + S'S \end{array}\right]$$

$$= \left[\begin{array}{c|c} 2^{n+1} I & 0 \\ \hline 0 & 2^{n+1} I \end{array}\right] = 2^{n+1} I_{2^{n+1}}.$$

H is Hadamard equivalent to V, and the proof is complete by Properties 4.1.2.

□

4.4 Sign matrices under the Cayley-Dickson Multiplication

The Cayley-Dickson doubling process is given by

$$(a, b)(c, d) = (ac - \overline{d}b, da + b\overline{c}). \tag{4.8}$$

When restricted to the frame of the 2^n-ons, the multiplication in (4.8) reduces to the form

$$\begin{aligned}
(\mathrm{a},\ 0)(\mathrm{c},0) &= (\mathrm{ac},\ 0)\\
(\mathrm{a},\ 0)(0,\mathrm{d}) &= (0,\ \mathrm{da})\\
(0,\ \mathrm{b})(\mathrm{c},0) &= (0, b\bar{c})\\
(0,\ \mathrm{b})(0,\mathrm{d}) &= -(\bar{d}b, 0)\ .
\end{aligned} \tag{4.9}$$

The conjugation is given by $\overline{(a,b)} = (\bar{a}, -b)$. We now consider the sign matrices of the corresponding frame multiplications.

4.4.1 Complex Numbers

The sign matrix is

$$S_{\mathbb{C}} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad \text{with} \quad S_{\mathbb{S}} S'_{\mathbb{S}} = 2I. \tag{4.10}$$

4.4.2 Quaternions

The frame multiplication is represented by Figure 4.3

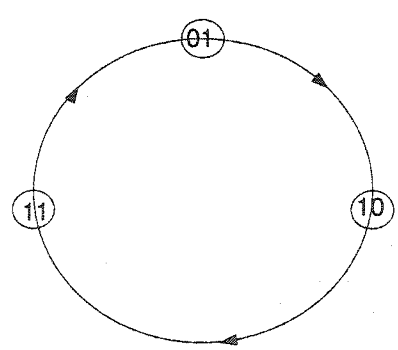

Figure 4.3 Cayley Dickson multiplication of quaternion frame

The sign matrix is

$$S_{\mathbb{H}} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & -1 \end{bmatrix} \quad \text{with} \quad S_{\mathbb{H}} S'_{\mathbb{H}} = 4I. \tag{4.11}$$

4.4.3 Octonions

The frame multiplication is represented by the Fano plane, Figure 4.4

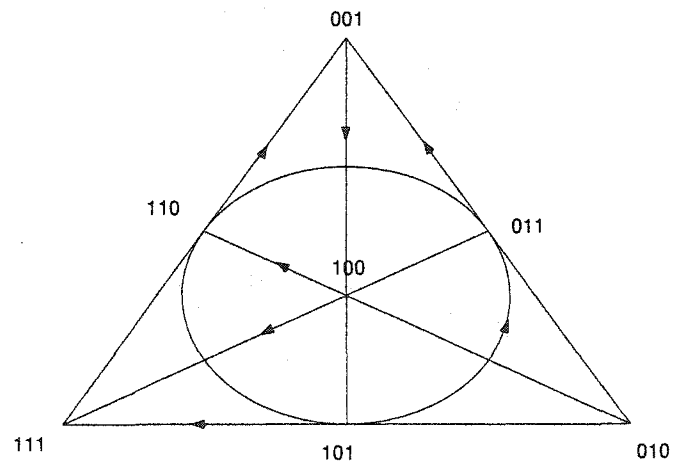

Figure 4.4 Cayley-Dickson multiplication of octonion frame

The sign matrix is

$$S_{\mathbb{K}} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 & -1 & 1 \\ 1 & -1 & -1 & 1 & 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & -1 & -1 & 1 & 1 & 1 \\ 1 & 1 & -1 & 1 & -1 & -1 & -1 & 1 \\ 1 & 1 & 1 & -1 & -1 & 1 & -1 & -1 \\ 1 & -1 & 1 & 1 & -1 & -1 & 1 & -1 \end{bmatrix} \quad \text{with} \quad S_{\mathbb{K}} S'_{\mathbb{K}} = 8I. \tag{4.12}$$

We proceed to show the following:

Theorem 4.4.1 *The sign matrix for the frame of the 2^n-ons under the Cayley-Dickson multiplication is a $2^n \times 2^n$ Hadamard matrix.*

Proof.

Let R, T be the sign matrices for the basic elements in the 2^n-ons and 2^{n+1}-ons respectively under the Cayley-Dickson multiplication. Let $\{e_0 = 1,\ e_1, \ldots,\ e_{2^n-1}\}$ be the basic elements of the 2^n-ons, then

$$\{(e_0, 0),\ (e_1, 0), \ldots,\ (e_{2^n-1}, 0),\ (0, e_0),\ (0, e_1), \ldots,\ (0, e_{2^n-1}\}$$

are the basic elements of the 2^{n+1}-ons.

We assume R is a $2^n \times 2^n$ Hadamard matrix and show that T is a $2^{n+1} \times 2^{n+1}$ Hadamard matrix. The matrix T is of the form

$$T = \left[\begin{array}{c|c} \mathrm{A} & \mathrm{B} \\ \hline \mathrm{C} & \mathrm{D} \end{array}\right], \tag{4.13}$$

where A, B, C and D are $2^n \times 2^n$ matrices specified as follows:

For $i, j > 0$

$$\begin{array}{rclll}
(e_{i-1}, 0)(e_{j-1}, 0) = (e_{i-1}e_{j-1}, 0) & \Rightarrow & A_{ij} = R_{ij}, & A = R & \\
(e_{i-1}, 0)(0, e_{j-1}) = (0, e_{j-1}e_{i-1}) & \Rightarrow & B_{ij} = R_{ji}, & B = R' & \\
(0, e_{i-1})(e_{j-1}, 0) = (0, e_{i-1}\overline{e_{j-1}}) & \Rightarrow & C_{i1} = R_{i1} = 1, & C_{ij} = -R_{ij} & j \neq 1 \\
(0, e_{i-1})(0, e_{j-1}) = (-\overline{e_{j-1}}e_{i-1}, 0) & \Rightarrow & D_{i1} = -R_{1i} = -1, & D_{ij} = -R_{ji} & i \neq 1.
\end{array}$$

Thus,

$$T = \left[\begin{array}{cccccc|cccccc}
1 & 1 & . & . & . & 1 & 1 & 1 & . & . & . & 1 \\
. & & & & & & . & & & & & \\
. & & & R_{ij} & & & . & & & R_{ji} & & \\
. & & & & & & . & & & & & \\
1 & & & & & & 1 & & & & & \\
\hline
1 & -1 & . & . & . & -1 & -1 & 1 & . & . & . & 1 \\
. & & & & & & . & & & & & \\
. & & & -R_{ij} & & & . & & & R_{ji} & & \\
. & & & & & & . & & & & & \\
1 & & & & & & -1 & & & & &
\end{array}\right].$$

We perform a column permutation $C_1 \leftrightarrow C_{2^n+1}$, i.e $C_{e_0,0)} \leftrightarrow C_{0,e_0}$, to get an equivalent matrix

$$H = \left[\begin{array}{cccccc|cccccc}
1 & 1 & . & . & . & 1 & 1 & 1 & . & . & . & 1 \\
. & & & & & & . & & & & & \\
. & & & R_{ij} & & & . & & & R_{ji} & & \\
. & & & & & & . & & & & & \\
1 & & & & & & 1 & & & & & \\
\hline
-1 & -1 & . & . & . & -1 & 1 & 1 & . & . & . & 1 \\
. & & & & & & . & & & & & \\
. & & & -R_{ij} & & & . & & & R_{ji} & & \\
. & & & & & & . & & & & & \\
-1 & & & & & & 1 & & & & &
\end{array}\right] = \left[\begin{array}{c|c} R & R' \\ \hline -R & R' \end{array}\right]. \quad (4.14)$$

Now,

$$HH' = \left[\begin{array}{c|c} R & R' \\ \hline -R & R' \end{array}\right]\left[\begin{array}{c|c} R' & -R' \\ \hline R & R \end{array}\right] = \left[\begin{array}{c|c} RR' + R'R & -RR' + R'R \\ \hline -RR' + R'R & RR' + R'R \end{array}\right]$$

$$= \left[\begin{array}{c|c} 2^{n+1}I & 0 \\ \hline 0 & 2^{n+1}I \end{array}\right] = 2^{n+1}I_{2^{n+1}}.$$

H is Hadamard equivalent to T, and the proof is complete by Properties 4.1.2

□

4.5 Skew Hadamard matrices

Definition 4.5.1 An Hadamard matrix of order m is called *skew* if

$$H = S + I \quad \text{and} \quad S' = -S.$$

If $H = S + I$ is a skew Hadamard matrix, the matrix S can be written in the form

$$S = \begin{bmatrix} 0 & e \\ -e' & W \end{bmatrix} \tag{4.15}$$

with e as an all-ones row vector. The matrix W in Equation (4.15) is called the *kernel* of the skew Hadamard matrix H. Yang [26] showed that

$$WW' = (m-1)I_{m-1} - J_{m-1} \quad \text{and} \quad WJ_{m-1} = 0, \quad W' = -W.$$

Proposition 4.5.2 *The sign matrix of the 2^n-ons frame under the Smith-Conway* (3.1) *or Cayley-Dickson* (4.9) *multiplication is equivalent to a skew Hadamard matrix.*

Proof.

The sign matrix of the frame under these two multiplications is of the form

$$\tilde{H} = \begin{bmatrix} 1 & & \cdots & & 1 \\ 1 & -1 & & \tilde{H}_{ij} & . \\ . & & . & & . \\ . & \tilde{H}_{ji} & & . & . \\ 1 & & \cdots & & -1 \end{bmatrix} \quad \text{with } \tilde{H}_{ji} = -\tilde{H}_{ij}. \tag{4.16}$$

If we multiply all rows by -1 except the first one we get an equivalent matrix

$$H = \begin{bmatrix} 1 & & \cdots & & 1 \\ -1 & 1 & & H_{ij} & . \\ . & & . & & . \\ . & H_{ji} & & . & . \\ -1 & & \cdots & & 1 \end{bmatrix} = I + \begin{bmatrix} 0 & 1 & \cdots & & 1 \\ -1 & 0 & & H_{ij} & . \\ . & & . & & . \\ . & H_{ji} & & . & . \\ -1 & & \cdots & & 0 \end{bmatrix} = I + S, \tag{4.17}$$

where $H_{ji} = -H_{ij}$ and so

$$S' = \begin{bmatrix} 0 & -1 & \cdots & & -1 \\ 1 & 0 & & H_{ji} & . \\ . & & . & & . \\ . & H_{ij} & & . & . \\ 1 & & \cdots & & 0 \end{bmatrix} = -S \quad \text{and } S = \begin{bmatrix} 0 & e \\ -e' & W \end{bmatrix}.$$

Here, W is the kernel of the sign matrix. $\square$

4.6 Sign Matrices and Kronecker products

Definition 4.6.1 Let $A = [a_{ij}]$ be an $n \times m$ matrix and let $B = [b_{ij}]$ be an $r \times s$ matrix. The *Kronecker product* of A and B is the $nk \times ms$ matrix given by

$$A \otimes B = \begin{bmatrix} a_{11}B & \cdots & a_{1m}B \\ \cdot \quad \cdot & & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ a_{n1}B & \cdots & a_{nm}B \end{bmatrix}. \tag{4.18}$$

The Kronecker product is also known as the *direct product* or the *tensor product.*

The following are some fundamental properties of Kronecker products.

Properties 4.6.2 *[10; 14]*

1. *The product is bilinear. Let* A, B and C *be matrices such that* B and C *have the same dimension and let* λ, β *be scalars, then:*

   (i) $A \otimes (B + C) = A \otimes B + A \otimes C$;

   (ii) $(B + C) \otimes A = B \otimes A + C \otimes A$;

   (iii) $\lambda(A \otimes B) = (\lambda A) \otimes B = A \otimes (\lambda B)$;

   (iv) $\lambda \otimes A = A \otimes \lambda = \lambda A$;

   (v) $\lambda A \otimes \beta B = \lambda\beta(A \otimes B)$.

2. *The product is associative:* $A \otimes (B \otimes C) = (A \otimes B) \otimes C$.

3. *If* A, B, C, D *are square matrices such that* AC *and* BD *exist, then*

$$(A \otimes B)(C \otimes D) = AC \otimes BD$$

4. *If* A and B *are invertible matrices, then* $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$.

5. *If* A and B *are square matrices, we have* $(A \otimes B)' = A' \otimes B'$.

6. *Let* A and B *be square matrices of dimension* n and m.

   Eigenvalues: *If* $\{\lambda_i | i = 1, \ldots n\}$ *are the eigenvalues of* A *and* $\{\mu_j | j = 1, \ldots m\}$ *are the eigenvalues of* B, *then* $\{\lambda_i \mu_j | i = 1, \ldots n, j = 1, \ldots m\}$ *are the eigenvalues of* $A \otimes B$.

   Determinant: $\det(A \otimes B) = (\det A)^n (\det B)^m$.

   Rank: $\operatorname{rank}(A \otimes B) = \operatorname{rank} A \cdot \operatorname{rank} B$.

   Trace: $\operatorname{trace}(A \otimes B) = \operatorname{trace} A \cdot \operatorname{trace} B$.

7. *For partitioned matrices,* $[A_1, A_2] \otimes B = [A_1 \otimes B, A_2 \otimes B]$, *but* $A \otimes [B_1, B_2] \neq [A \otimes B_1, A \otimes B_2]$ *in general.*

Let H_1 and H_2 be Hadamard matrices of order n and m respectively. The Kronecker product $H_1 \otimes H_2$ is a Hadamard matrix of order nm. This follows easily from Properties 4.6.2 since

$$(H_1 \otimes H_2)(H_1 \otimes H_2)' = (H_1 \otimes H_2)(H_1' \otimes H_2')$$

$$= H_1 H_1' \otimes H_2 H_2' = nI_n \otimes mI_m = nmI_{nm}.$$

The Kronecker product of the sign matrix for the complex numbers is given by

$$S_{\mathbb{C}} \otimes S_{\mathbb{C}} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}.$$

This matrix is equivalent to the sign matrix of the quaternions $S_{\mathbb{H}}$ in Equation (4.3) via the row permutation $R_2 \leftrightarrow R_4$ or the column permutation $C_3 \leftrightarrow C_4$. The Kronecker product of the sign matrices of the complex numbers and quaternions is given by

$$S_{\mathbb{C}} \otimes S_{\mathbb{H}} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 \\ 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \end{bmatrix},$$

and is equivalent to the sign matrix of the octonions given by (4.4). The equivalence of $S_{\mathbb{K}}$ and the Kronecker product $S_{\mathbb{C}} \otimes S_{\mathbb{H}}$ is achieved via the permutation

$$R_2 \to R_7 \to R_3 \to R_8 \to R_4 \to R_6 \to R_2.$$

If we start with the Kronecker product $S_{\mathbb{H}} \otimes S_{\mathbb{C}}$, we get an equivalent matrix to $S_{\mathbb{K}}$. This time we use the row permutation

$$R_2 \to R_6 \to R_4 \to R_2, \quad R_7 \to R_8 \to R_7.$$

We note that $(S_{\mathbb{C}} \otimes S_{\mathbb{C}}) \otimes S_{\mathbb{C}} = S_{\mathbb{C}} \otimes (S_{\mathbb{C}} \otimes S_{\mathbb{C}})$ is also equivalent to the matrix $S_{\mathbb{K}}$ with the permutation given by

$$R_2 \to R_6 \to R_4 \to R_7 \to R_3 \to R_8 \to R_2.$$

The immediate question is:

Is the Kronecker product of the sign matrix of the 2^{n-1}-ons with that of $S_{\mathbb{C}}$ always equivalent to the sign matrix of the 2^n-ons?

If S is the sign matrix of the 2^{n-1}-ons, the sign matrix of the 2^n-ons is equivalent to the matrix

$$H = \left[\begin{array}{c|c} S & S' \\ \hline -S & S' \end{array}\right].$$

as in (4.7). On the other hand, the Kronecker product

$$S_{\mathbb{C}} \otimes R = \left[\begin{array}{c|c} R & R \\ \hline R & -R \end{array}\right].$$

The problem would be solved if these two matrices were shown to be equivalent. The problem is still open.

CHAPTER 5. FUTURE RESEARCH PROBLEMS

1. In this research work we gave a way of counting the number of frame subloops of order 2^k. The problem of classifying all the subloops of the various 2^n-ons remain open.

2. The problem of determining the equivalence between the sign matrix of the frame of the 2^n-ons and the Kronecker product of the sign matrix of the 2^{n-1} and that of the complex numbers is open for $n \geq 4$. One can approach this problem using the character tables of the frames.

3. Take any normalized Hadamard matrix of order 2^n, use it as a sign matrix together with the projective geometry to define an algebra. What are the properties of the algebra? If one uses two equivalent matrices, how are the algebras related? Are they isotopic or isomorphic?

4. Hadamard matrices give a frame that is a loop. What are the properties of the loop?

5. In Lemma 2.3.1, each commutative multiplicative loop of the octonions is associative. Corollary 2.4.6 shows that each commutative sedenion extension loop is associative, this Corollary would be a more complete analogue of Lemma 2.3.1 if it were known that the only two-sided subloops of the multiplicative left loop of non-zero sedenions appear either as subloops of the octonions, or as sedenion extensions.

6. In Chapter 2, the equivalent properties for the sedenion extension loop to be a group were discussed. The characterization of the "arbitrary" sedenion extension remains open. Of interest is the properties of extensions formed by loops other than the subloops in $\mathbb{K}$.

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